Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

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Abstract

The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone. This study examines the relationship between wall-actuation structure, hydrodynamic response, and cycle-mean viscous dissipation in a two-dimensional lid-driven cavity. Stationary harmonic profiles and three physics-informed actuation branches are transferred to an independent multiple-relaxation-time lattice-Boltzmann solver and compared at a common quadratic actuation norm. For stationary controls at $Re = 500$ on the $N = 512$ lattice, the first spatial harmonic reduces dissipation by 17.7% relative to a corner-regularized uniform-lid reference, whereas the second harmonic and the mean profile of the quasi-steady two-lobe branch increase it by 36.8% and 44.3%, respectively. Grid-refinement and local strain-rate analyses associate this ordering with the redistribution of dissipation between corner shear layers and the cavity interior. For time-dependent branches, the cycle-mean dissipation differs markedly from the dissipation of the time-mean flow: the fluctuating part of the strain field carries 92.9%, 53.7% and 1.8% of the cycle-mean cost of the lid-oscillatory, mixed and quasi-steady branches, respectively. The branch for which the dissipation evaluated on the time-mean field is 14.2 times smaller than the cycle-mean dissipation is consequently the most dissipative of the three, though by the modest factor of 1.2, while the quasi-steady and mixed responses differ by 2.1% at $N = 256$ and by 0.5% at $N = 512$. A second-Stokes boundary-layer estimate built from the imposed lid signal alone agrees with the measured fluctuating dissipation of the lid-oscillatory and mixed branches to within approximately 5% and 2%, respectively, without an adjustable constant. The estimate is a half-space approximation, so this agreement supports a wall-layer interpretation of the dominant fluctuating cost rather than a spatial decomposition of the total dissipation. The same balance explains why the quasi-steady and mixed branches, whose oscillatory actuation energies differ by a factor of 36, cost the same over the cycle to within 2.1%: their totals are reached through opposite partitions between mean field and fluctuations. The results demonstrate that the energetic cost of wall actuation depends on both the spatial organization and the temporal fluctuations of the resulting flow, and that physical performance must be assessed using cycle-averaged dissipation rather than modal accessibility or mean-field diagnostics alone.

Keywords: lid-driven cavity; wall actuation; viscous dissipation; energy budget; mean-flow diagnostics; mode selection

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Nomenclature

Roman symbols	
A_ε	half-spread of $D(t)$ over the cycle, $A_\varepsilon = (D_{\max} - D_{\min})/2$, i.e. the variability concealed by a single cycle-mean number
AR	cavity aspect ratio, L_x/L_y
C_3	spectral rank concentration, sum of the three largest control-energy fractions
D	instantaneous viscous dissipation rate, $(2/Re)\int_\Omega \mathbf{S}:\mathbf{S}\, d\Omega$
$\overline{D}$	cycle-mean dissipation, $\langle D\rangle_t$
E_{act}	instantaneous quadratic actuation norm of the lid profile, $\int_0^1 U_{\text{lid}}^2\, dx$
E_r	realized cycle-averaged actuation energy of a transferred control
E^*	prescribed target actuation energy
f_1, f_2, f_t	first-harmonic, second-harmonic, and explicitly time-dependent control-energy fractions
H_N	normalized branch-selection entropy
K	kinetic energy, $\frac{1}{2}\int_\Omega \lvert\mathbf{u}\rvert^2\, d\Omega$
L, L_x, L_y	cavity side, streamwise and wall-normal lengths
N	number of lattice nodes per cavity side (LBM resolution)
P_{lid}	mechanical power injected through the lid, $\int_0^1 U_{\text{lid}}\tau_w\, dx$
R_K	fraction of the cycle-mean kinetic energy carried by temporal fluctuations, $\overline{K}_{\text{fluct}}/\overline{K}$ with $\overline{K} = \overline{K}_{\text{mean}} + \overline{K}_{\text{fluct}}$
Re	Reynolds number based on the physical scaling velocity, $Re = U_0 L/\nu$; in the LBM mapping, U_0 is represented by $U_{\text{ref}} = 0.05$ in lattice units
$\mathbf{S}$	strain-rate tensor, $(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathsf{T}})/2$
t, T	time and forcing period
$\mathbf{u} = (u, v)$	velocity vector and its streamwise/wall-normal components
$\mathbf{u}'$	fluctuating velocity vector, $\mathbf{u} - \langle\mathbf{u}\rangle$
u'_{rms}	root-mean-square temporal fluctuation at a fixed point
U_{lid}	lid velocity profile (the actuation)
U_0	physical scaling velocity of the nondimensional problem, $Re = U_0 L/\nu$; the uniform lid at $E^* = 0.25$ has amplitude $U_{\text{uniform}} = 0.5\, U_0$
U_{ref}	lattice reference velocity of the MRT-LBM solver, 0.05 in lattice units; it plays the role of U_0 in the lattice nondimensionalisation and sets $\Delta t = U_{\text{ref}}/N$
x, y	streamwise and wall-normal coordinates
Greek symbols and branch symbols	
B_1, B_2, B_3	empirical branch classes: quasi-steady two-lobe, lid-oscillatory, mixed
B_{first}	first-harmonic branch class of the full-spectrum audit
δ	corner taper width
$\varepsilon(t; N)$	instantaneous viscous dissipation rate integrated over the cavity domain at lattice resolution N; its cycle average is $\overline{\varepsilon}$
λ_{var}	weight of the variance-regularization term of the training loss
ν	kinematic viscosity
τ_w	wall shear stress on the moving lid
ψ	stream function
Ω	cavity domain
Abbreviations	
GCI	grid-convergence index
LBM	lattice-Boltzmann method
MRT	multiple-relaxation-time
PINN	physics-informed neural network
RMSE	root-mean-square error

1 Introduction

Wall actuation is one of the few ways in which the vortical structure of a confined recirculating flow can be modified without changing its geometry. In a lid-driven cavity the entire vortex topology is generated by the tangential motion of a single boundary, so any spatial structure imposed on that boundary acts directly on the mechanism that sustains the flow. The cavity is therefore a natural configuration in which to ask a question that is central to flow control but

rarely posed in energetic terms: when two different actuations require the same control effort, which one produces the less dissipative flow, and why? The canonical cavity solution and its benchmarks are well established [1, 2, 3, 4, 5], and the flow has long served as a test bed for control strategies [11, 12, 13, 14, 15, 16, 17, 19, 20], yet its dissipation budget under distributed wall actuation has received comparatively little attention.

The energetic mechanism is twofold, and the two parts can act in opposite directions. Changing the wall velocity changes the mechanical power injected into the fluid, $P_{\mathrm{lid}} = \int_0^1 U_{\mathrm{lid}} \tau_w \, dx$, where τ_w is the wall shear stress; it also changes the spatial distribution of the strain-rate tensor $\mathbf{S}$, and with it the partition of viscous dissipation between the lid shear layer, the side-wall layers, the cavity interior and the two upper corners, where a discontinuous lid produces a stress singularity. A profile can therefore reduce the peak strain at the corners while increasing the strain everywhere else, and the net effect is not predictable from the shape of the actuation alone. The distinction matters because the two natural comparison protocols answer different questions. Comparing profiles at the same lid-velocity scale [38] measures what a given actuator amplitude achieves; comparing them at the same quadratic actuation norm $E_{\mathrm{act}} = \int_0^1 U_{\mathrm{lid}}^2 \, dx$ measures what a given control effort achieves. The two rankings need not agree, and only the second is a statement about the energetic response at fixed actuation norm. A further distinction, specific to time-dependent actuation, is that the dissipation of the time-mean field is not the time mean of the instantaneous dissipation: because D is quadratic in the velocity gradients, the cycle-averaged dissipation is no smaller than the dissipation of the time-mean field, with a strict inequality whenever the fluctuating strain-rate field is nonzero, and the gap can be orders of magnitude when the fluctuations dominate [28, 29, 30, 27].

Spatially structured and time-dependent wall actuations can be recovered in distinct modal families by the identification procedure, and the family that is selected depends on the formulation, the regularization and the initialization of the optimization. These families are distinct branches of the control-identification landscape; they are not evidence of several hydrodynamic states coexisting under one and the same prescribed lid motion. Each exported actuation is therefore treated here as a separate boundary forcing, and its periodic flow response is computed independently with the MRT-LBM solver. A preceding study by Beniaiche [38] investigated the identification of conditional modal collapse in physics-informed active control of the lid-driven cavity, building on modal decomposition and reduced-order modeling techniques [24, 25, 26, 23]. It established that the identified actuation is not characterized by a unique, universally selected modal structure: predominantly stationary second-harmonic controls and strongly time-dependent alternatives can be recovered depending on initialization, regularization, and control parameterization. In that work, as here, the network identifies candidate spatio-temporal wall-actuation profiles, which are then evaluated with an independent MRT-LBM solver; the solver is used to compute the response to a profile, not to validate the network. The emphasis of that work was on branch identification, conditional accessibility, and independent reproducibility of the responses. The present study addresses the complementary fluid-mechanical question of how these spatial and temporal actuation structures affect the viscous-dissipation budget of the resulting flow. Rather than treating modal recovery as a performance criterion, we quantify the dissipation of stationary and periodic responses under a common quadratic actuation norm, distinguish the dissipation of the time-mean field from the cycle-mean dissipation, and identify the strain-rate distributions responsible for the observed energetic differences.

The energetic question is not the one usually asked of an oscillating cavity. Cavities driven by a periodically moving lid have been studied for the patterns and the transport they produce: the lattice-Boltzmann computation of a cavity driven by an oscillating lid by Mendu and Das [6], the vortex dynamics and flow-pattern map of the oscillatory lid-driven rectangular cavity by Zhu et al. [7], Lagrangian transport in the time-periodic two-dimensional lid-driven square cavity by Babor and Kuhlmann [8], and the chaotic mixing of suspensions in a time-periodic lid-driven cavity by Xu and Gilchrist [9]. In each case the figure of merit is mixing, pattern selection or material transport. The dissipation that the actuation costs at a prescribed control energy is

a different figure of merit—the one that governs actuator sizing and that an energy-respecting control or reduced-order model must reproduce—and it is the quantity computed here. Likewise, the gap between the dissipation of a mean field and the cycle mean of a quadratic functional is classical, the boundary-layer theory of oscillating walls is classical, and the steady streaming that the Reynolds stresses of the fluctuations drive is a subject with its own literature [10]. What was missing, for this geometry, is the connection between the three: a measured dissipation cost at a common actuation norm, an estimate of the temporal part of that cost from the imposed lid signal alone, and a field-level account of why the mean field is a poor proxy for it.

Three statements mark the novelty of the present contribution, and none of them is Jensen's inequality. First, the ranking of stationary actuations is shown to be a property of the constraint, not of the flow: the same three profiles are ordered differently at equal lid speed and at equal quadratic norm, and the inversion is associated with the change in peak velocity and hence in effective Reynolds number of each profile (Sec. 5). Second, the part of the cycle mean that the mean field misses is set by the oscillating Stokes layer: the second-Stokes scaling of the imposed lid harmonics agrees with the measured fluctuating dissipation of the lid-oscillatory and mixed branches to within approximately 5% and 2%, respectively, and because all three controls place nearly all of their oscillatory energy in the first temporal harmonic, the estimate collapses to the one-line form $D_{\mathrm{fluct}} \simeq f_t \overline{E}_{\mathrm{act}} \sqrt{\nu\omega_1/2}$. This turns an empirical ranking into a boundary-layer statement that can be checked without solving the field problem (Sec. 6.3). Third, the weak mean field of the oscillatory branch is consistent with a Reynolds-stress-driven steady-streaming mechanism, so the failure of mean-field diagnostics has a plausible dynamical explanation and not only an arithmetic one (Sec. 6.4). Together the three statements have an energetic consequence that is the central result of this paper: at a matched actuation norm, three controls with very different structure—quasi-stationary, mixed and essentially oscillatory—dissipate within a narrow band, $\overline{D} = 1.19 \times 10^{-2}$, 1.17×10^{-2} and 1.44×10^{-2}, and the oscillatory branch that suppresses its mean flow most effectively pays for it almost exactly in fluctuations.

The question addressed in this paper is accordingly:

> How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?

The sequence of the paper follows that question. Section 2 fixes the physical and energetic framework, and in particular separates the quadratic actuation norm from the mechanical power transmitted through the wall. Section 3 defines the stationary modes and the three identified branches that are compared. Section 4 describes how the actuations are identified and how the flows are computed independently. Section 5 reports the energetic response of the stationary modes, Section 6 the cycle-mean response of the time-dependent branches, and Section 7 interprets both in terms of local strain and fluctuating gradients. Section 8 concludes.

2 Physical formulation and energetic framework

2.1 Controlled cavity and governing equations

We consider a two-dimensional incompressible flow in the square cavity $0 \le x \le 1$, $0 \le y \le 1$, made dimensionless with the cavity height L and a physical scaling velocity U_0 (Fig. 1a). The lower and lateral walls are no-slip and stationary; the upper wall moves tangentially with a velocity profile $U_{\mathrm{lid}}(x,t)$ that is the control variable. The velocity field $\mathbf{u} = (u, v)$ and the pressure p satisfy

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2\mathbf{u}, \qquad \nabla\cdot\mathbf{u} = 0, \tag{1}$$

with $Re = U_0 L/\nu$, subject to

$$\mathbf{u} = \mathbf{0} \ \text{ on } x = 0,\ x = 1,\ y = 0, \qquad u = U_{\mathrm{lid}}(x,t),\ \ v = 0 \ \text{ on } y = 1. \tag{2}$$

A pointwise tangential velocity along the lid is an idealized boundary condition for distributed tangential control. It may represent an array of independently driven wall segments or approximate the time-averaged effect of a distributed momentum source such as a dielectric-barrier-discharge actuator [18]; the present formulation does not assume a detailed physical equivalence with any particular actuation technology. Because U_0 is a scaling velocity and not a property of the lid, a profile may change sign along x or in time; the amplitude of the actuation is fixed separately by the norm constraint introduced in Sec. 2.2.

2.2 The quadratic actuation norm and the mechanical wall power

Two different scalar quantities characterize a wall actuation, and they must not be conflated. The first is the *quadratic actuation norm*

$$E_{\mathrm{act}}(t) = \int_0^1 U_{\mathrm{lid}}^2(x,t)\,dx, \qquad \overline{E}_{\mathrm{act}} = \frac{1}{T}\int_0^T E_{\mathrm{act}}(t)\,dt, \tag{3}$$

which measures the squared amplitude of the commanded wall velocity, is dimensionless in the scaling of Eq. (1) and corresponds to $U_0^2 L$ in dimensional variables, and is the quantity held fixed in all comparisons below. It is a norm of the boundary data, not a work: it contains no information about the fluid response. The second is the *mechanical power* transmitted by the lid,

$$P_{\mathrm{lid}}(t) = \int_0^1 U_{\mathrm{lid}}(x,t)\,\tau_w(x,t)\,dx, \qquad \tau_w = \frac{1}{Re}\left.\frac{\partial u}{\partial y}\right|_{y=1}, \tag{4}$$

positive when the wall does work on the fluid. Here U_{lid} is the prescribed wall velocity. In the numerical evaluation (Table 4 and Sec. 6.5), the power is estimated using the recovered wall velocity $U_{\mathrm{wall}}^{\mathrm{rec}}$ and a finite-difference wall shear τ_w^{FD}:

$$P_{\mathrm{lid}}^{\mathrm{rec}}(t) = \int_0^1 U_{\mathrm{wall}}^{\mathrm{rec}}(x,t)\,\tau_w^{\mathrm{FD}}(x,t)\,dx, \tag{5}$$

with $U_{\mathrm{wall}}^{\mathrm{rec}}(x_j,t) = u_x(N-1,j)$ the streamwise velocity of the lid node row and $\tau_w^{\mathrm{FD}}(x_j) = Re^{-1}[u_x(N-1,j) - u_x(N-2,j)]/\Delta y$ the one-sided wall-normal difference between that row and the interior row immediately beneath it, Δy being the node spacing of the archived fields (Sec. 4.2, Supplementary Sec. S4.8). This distinction is necessary because the half-way boundary condition imposes the commanded velocity at the hydrodynamic wall, the plane a half spacing outside the lid node row, so the velocity of that row is a computed quantity rather than the prescribed one; the recovered velocity estimator gives cycle-mean residuals of -2.8%, -0.6% and -0.4% for branches B_1, B_2 and B_3, whereas using the prescribed velocity directly gives residuals of $+3.4\%$, $+4.6\%$ and $+4.7\%$ [on the phase-resolved fields, against -2.9%, -6.0% and -4.3% for the recovered one]. The gap between those two sets is a discrepancy of the discrete power budget to which the near-wall nodal velocity offset — the difference between the velocity of the lid node row and the commanded profile — contributes; it is not a direct measurement of that offset, nor of the continuous energy balance (Sec. 6.5). We note that $P_{\mathrm{lid}}^{\mathrm{rec}}$ is a diagnostic of the power across the *last fluid row of the lattice*, the node row $i = N-1$ that carries the moving lid in the array layout, with the wall-normal derivative taken as the one-sided difference between rows $N-1$ and $N-2$ (Supplementary Sec. S4.8). Because the boundary condition is a half-way moving-wall rebounce with momentum injection, the hydrodynamic wall lies half a lattice spacing outside that row, so the row is a first-plane approximation of the wall rather than the wall itself: it tends to the wall power P_{lid} of Eq. (4) as $\Delta y \to 0$, and it also explains why the velocity recovered at that row differs from the one commanded at the wall plane by the near-wall nodal velocity offset measured in Supplementary Sec. S4.8, whose largest value over the branch records spans 3.2% to 11.8% of the cavity speed as the grid goes from $N = 512$ to $N = 256$. That difference is a consequence of where the velocity is sampled, half a spacing inside the hydrodynamic wall, and is not a physical slip;

it is expected to decrease under grid refinement, which Sec. 6.5 verifies. The residuals reported above therefore verify the discrete energy budget indirectly rather than close it exactly at the boundary, and are consistent with the exact identity $\overline{P_{\mathrm{lid}}} = \overline{D}$ of the continuous equations. P_{lid} depends on the actuation and on the flow through the wall shear stress, and it is the quantity that enters the energy balance. Two actuations of equal $\overline{E}_{\mathrm{act}}$ can therefore inject very different amounts of power; conversely, a large norm does not guarantee a large power if the wall shear is weak.

Holding $\overline{E}_{\mathrm{act}}$ fixed also fixes the root-mean-square wall velocity, $\overline{E}_{\mathrm{act}}^{1/2} = 0.5$ at the value $E^* = 0.25$ used throughout, but not the maximum wall velocity. A uniform lid of unit shape then has amplitude 0.5, whereas a first harmonic of the same norm has amplitude $\sqrt{2E^*} = 0.7071$; a peakier profile has a larger maximum still. Because the viscosity is fixed, a profile whose peak wall velocity is $U_{\max}$ realizes an effective Reynolds number $Re_{\max} = U_{\max} Re$ based on that peak. The comparisons below are therefore comparisons at equal quadratic actuation norm, at fixed viscosity and geometry, and *not* at equal $Re_{\max}$; the values of $U_{\max}$ and $Re_{\max}$ are tabulated for every case (Tables 1 and 2) so that the distinction is explicit at every step. This convention differs from the fixed-lid-velocity comparison of Ref. [38], in which the peak velocity was the same for all profiles and the norm differed by up to a factor of four.

2.3 Viscous dissipation and its cycle mean

For an incompressible Newtonian fluid the instantaneous viscous dissipation rate is

$$D(t) = \frac{2}{Re} \int_\Omega \mathbf{S} : \mathbf{S} \, d\Omega, \qquad \mathbf{S} = \tfrac{1}{2} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right), \tag{6}$$

and the quantity relevant to a periodic actuation is its cycle average

$$\overline{D} = \frac{1}{T} \int_0^T D(t) \, dt. \tag{7}$$

Splitting the velocity into a time mean and a fluctuation, $\mathbf{u} = \overline{\mathbf{u}} + \mathbf{u}'$ with $\mathbf{S} = \overline{\mathbf{S}} + \mathbf{S}'$, the cycle mean decomposes exactly as

$$\overline{D} = \underbrace{\frac{2}{Re} \int_\Omega \overline{\mathbf{S}} : \overline{\mathbf{S}} \, d\Omega}_{D(\overline{\mathbf{u}})} + \underbrace{\frac{2}{Re} \left\langle \int_\Omega \mathbf{S}' : \mathbf{S}' \, d\Omega \right\rangle}_{\text{fluctuating part}}, \tag{8}$$

because the cross term vanishes upon averaging. The second term is a cycle-averaged squared norm, so it is non-negative and vanishes only when the fluctuating strain-rate field is zero; the two terms therefore obey $\overline{D} \geq D(\overline{\mathbf{u}})$, with a strict inequality whenever the response fluctuates, and the gap measures how much of the dissipation is carried by the unsteady part of the response. Evaluating D on a time-averaged field is thus not an approximation of $\overline{D}$ with a small error: it is a different functional, and for a strongly unsteady flow it can invert the ranking of two actuations.

2.4 Mechanical energy budget

With the kinetic energy $K = \frac{1}{2} \int_\Omega |\mathbf{u}|^2 \, d\Omega$, the momentum equation gives

$$\frac{dK}{dt} = P_{\mathrm{lid}}(t) - D(t). \tag{9}$$

For a periodic state the left-hand side integrates to zero over one period, so

$$\boxed{\overline{P}_{\mathrm{lid}} = \overline{D}} \tag{10}$$

is an exact identity of the governing equations. In the numerical evaluation of Eq. (10) both terms are computed from the same discrete fields, but P_{lid} involves a wall-normal derivative at the

boundary, which is evaluated with a one-sided finite difference and therefore carries a truncation error that D does not. The budget is used below as a verification of the transfer and of the periodicity of the response, with that estimator uncertainty made explicit (Sec. 6.5); it is not used as an exact closure constraint.

3 Actuation modes and hydrodynamic states

3.1 Stationary reference profiles

Five stationary profiles define the comparison, all normalized to $E^* = 0.25$. The *uniform* lid $U_{\mathrm{lid}} = 0.5$ is the classical reference; it carries a tangential-velocity discontinuity at the two upper corners, which is retained because it is the classical boundary condition, but which is shown in Sec. 5.2 to prevent the convergence of its own dissipation. A *corner-regularized uniform* lid, constructed with an initial plateau $U_p = 0.5$ and a raised-cosine taper of width $\delta = 0.05$ at each corner

$$U_{\mathrm{lid}}^{\mathrm{raw}}(x) = \begin{cases} \dfrac{U_p}{2}\left[1 - \cos\left(\dfrac{\pi x}{\delta}\right)\right], & 0 \le x < \delta, \\ U_p, & \delta \le x \le 1-\delta, \\ \dfrac{U_p}{2}\left[1 - \cos\left(\dfrac{\pi(1-x)}{\delta}\right)\right], & 1-\delta < x \le 1, \end{cases} \tag{11}$$

then rescaled to the prescribed norm:

$$U_{\mathrm{lid}}(x) = c\, U_{\mathrm{lid}}^{\mathrm{raw}}(x), \qquad c = \sqrt{\frac{E^*}{\displaystyle\int_0^1 (U_{\mathrm{lid}}^{\mathrm{raw}})^2\, dx}}. \tag{12}$$

For $\delta = 0.05$, the raw profile carries $E_{\mathrm{raw}}/E_{\mathrm{uniform}} = 1 - 5\delta/4 = 0.9375$, a 6.25% loss which is restored by the rescaling factor $c = \sqrt{16/15} \approx 1.033$; the final plateau value is approximately 0.5164, corresponding to $U_{\max} = 0.52$ in table 1. This regularized profile removes the discontinuity and provides a grid-convergent reference. The first and second spatial harmonics, $U_{\mathrm{lid}} = \sqrt{2E^*}\sin(\pi x)$ and $\sqrt{2E^*}\sin(2\pi x)$, are the two elementary spatially structured controls. Finally, the time-mean profile of the identified B_1 branch (Sec. 3.2) is included as the stationary approximation of a physics-informed control.

3.2 Identified actuation branches

The time-dependent actuations are three representatives identified by the physics-informed procedure of Sec. 4 and classified by the modal energy of their control coefficients (details of the identification campaign and classification rules are provided in Supplementary Secs. S1–S3, particularly Sec. S2.3 and Tables S2–S3). Writing the lid as

$$U_{\mathrm{lid}}(x,t) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t-1} a_{mn} \sin(m\pi x) \cos\left(\frac{2n\pi t}{T}\right), \tag{13}$$

with forcing period $T = 2$, the energy of coefficient (m,n) is $\mathcal{E}_{m0} = \frac{1}{2}a_{m0}^2$ and $\mathcal{E}_{mn} = \frac{1}{4}a_{mn}^2$ for $n \ge 1$, with normalized fractions $f_{mn} = \mathcal{E}_{mn}/\sum \mathcal{E}_{ij}$. The stationary second-harmonic fraction is $f_2 \equiv f_{20}$ and the total time-dependent fraction is $f_t = \sum_{m=1}^{N_x}\sum_{n=1}^{N_t-1} f_{mn}$. The three classes are defined by modal dominance [38]: B_1 has $f_2 > 2f_t$ and $f_2 > 0.5$ (stationary second-harmonic dominated); B_2 has $f_t > 2f_2$ and $f_t > 0.5$ (temporally dominated); B_3 satisfies neither condition (mixed). At $Re = 500$ and $E^* = 0.25$ the representatives used here have $(f_2, f_t) = (0.92, 0.013)$ for B_1 (the seedless reference Fourier export of the parent campaign; Supplementary Sec. S4.6 gives its archive and checksum), $(0.00, 1.00)$ for B_2 (seed 5) and $(0.52, 0.45)$ for B_3 (seed 0), with a first-harmonic content $f_1 = 0.006$ for B_3 and below 0.02 for the other two.

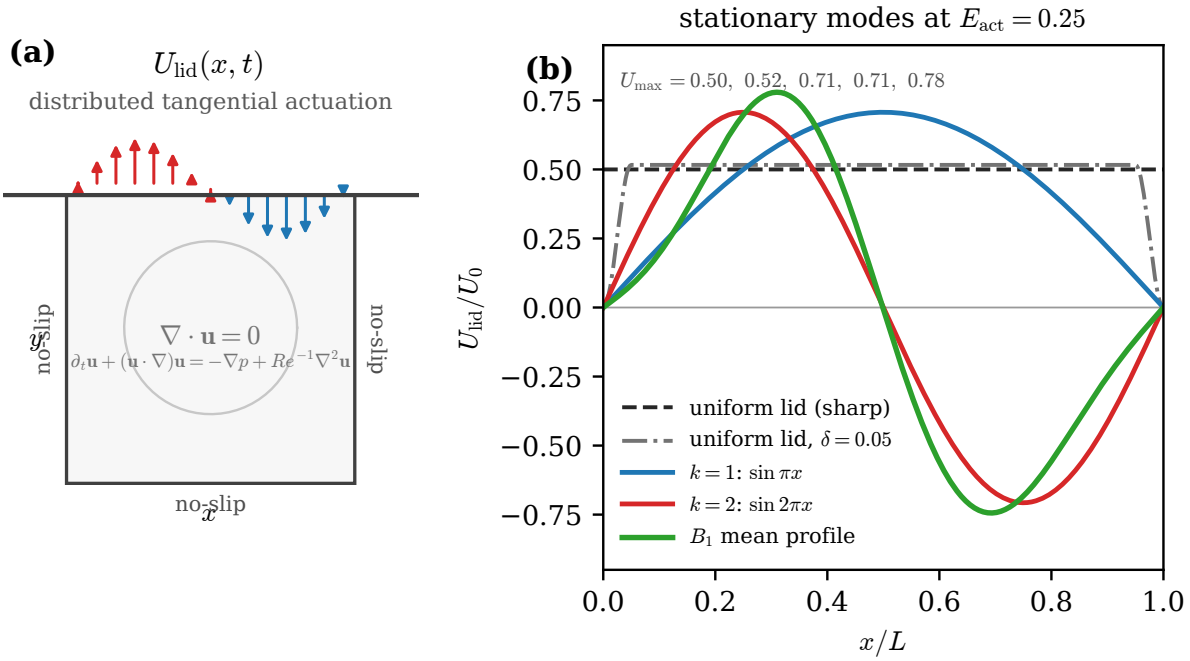


Figure 1: (a) Geometry of the controlled cavity. The upper wall moves tangentially with a distributed profile $U_{\mathrm{lid}}(x,t)$; the three other walls are no-slip and stationary. (b) The five stationary actuation profiles compared in this study, all normalized to the same quadratic actuation norm $E^*=\int_0^1 U_{\mathrm{lid}}^2\,dx=0.25$. The maximum wall velocity differs between profiles at equal norm: $U_{\max}=0.50$ for the sharp uniform lid, 0.52 for the corner-regularized uniform lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile.

Those rules are modal, but each class corresponds to a distinct wall motion, and stating it matters because the labels are only identifiers. Decomposing the three imposed lid series in the spatial sine basis, harmonic by harmonic (Supplementary Sec. S2.3), gives the following. The stationary part of B_1 holds 98.7% of its actuation energy and is 93% the second spatial harmonic ($a_{20}=0.68$): a two-lobe lid whose wall velocity keeps its sign over the whole span except at the midline node, where that profile vanishes and the residual oscillation sets the sign. B_2 puts 97.7% of its energy in the first temporal harmonic, and 98.5% of that harmonic in the single-lobe first spatial mode ($a_{11}=0.98$), so the whole wall reverses direction twice per period and its time mean is nearly zero. B_3 superposes the two: a stationary second-harmonic lid (55% of the energy, 94% of it in $m=2$) and an oscillating single-lobe lid (43%, 91% in $m=1$); the two cancel over the central band $0.28<x<0.64$, which is the 36% of the span that reverses. In words, then, the three classes are the *quasi-steady two-lobe* branch (B_1), the *lid-oscillatory* branch (B_2) and the *mixed* branch (B_3); the symbols are kept in the tables and figures, where they denote particular recovered controls rather than a physical type.

The B_1 representative is used in two forms. The *full* series is the control as identified, including its small temporal content; the *mean profile* is its time average, a stationary actuation that isolates the spatial structure. The stationary comparison of Sec. 5 uses the mean profile and the cycle-mean comparison of Sec. 6.2 uses the full series, so that the effect of the spatial structure is never conflated with the effect of the temporal one. Representative profiles and their space–time structure are shown in Fig. 2; Table 1 collects the characteristics of the eight primary actuation cases compared in the paper.

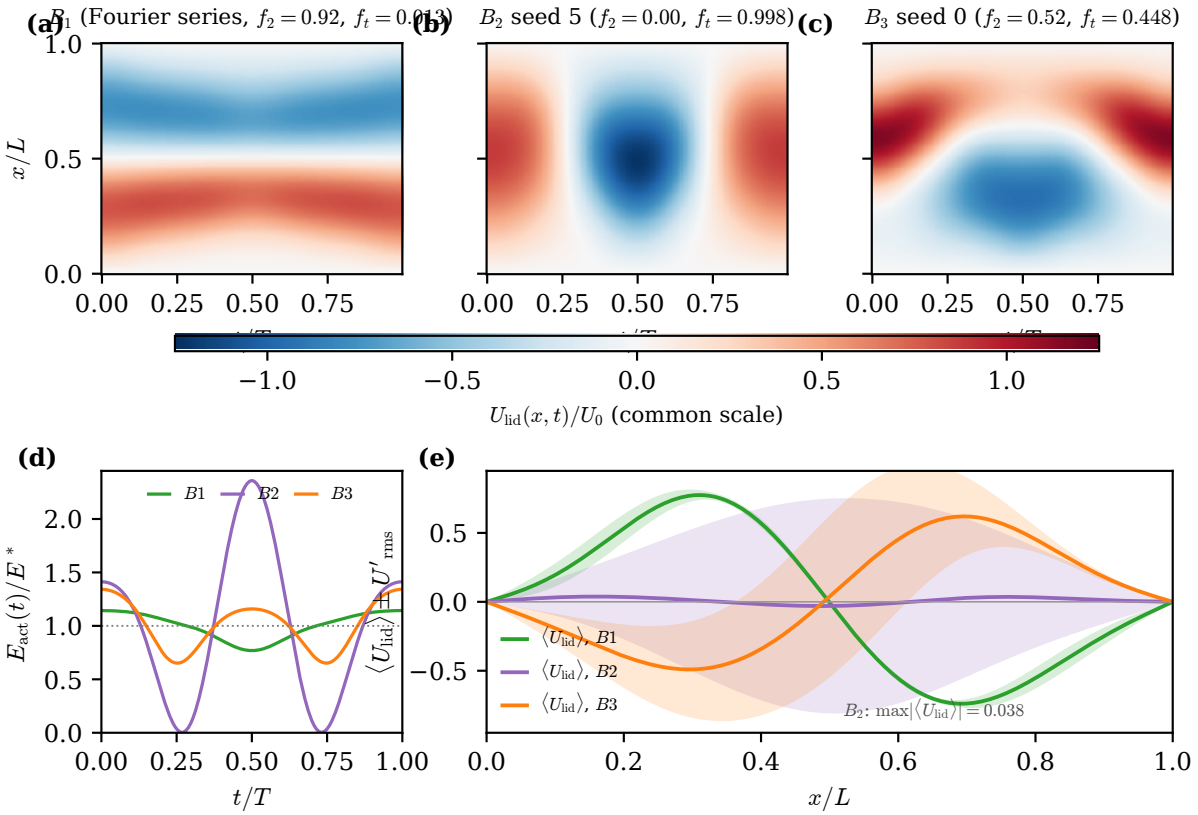


Figure 2: Spatio-temporal structure of the three identified branch controls at $Re = 500$ and $\overline{E}_{\mathrm{act}} = 0.25$. (a)–(c) Lid velocity $U_{\mathrm{lid}}(x,t)$ over one forcing period, on a common colour scale. B_1 is nearly stationary; B_2 oscillates with a full sign reversal, the wall velocity vanishing twice per period; B_3 combines a stationary two-lobe pattern with a strong oscillation. (d) Instantaneous actuation norm $E_{\mathrm{act}}(t)$ normalized by the common cycle-mean target: B_2 spans 0.004–2.36 E^*, whereas B_1 stays within 0.77–1.14 E^* and B_3 within 0.65–1.34 E^*. The comparison is therefore made at equal *cycle-mean* norm, not at equal instantaneous norm. (e) Time-mean lid profile with the temporal standard deviation as a band; the B_2 mean profile reaches only 0.038, so a mean-field diagnostic sees almost no actuation.

4 Numerical methodology

4.1 Identification of the actuation

The control coefficients $a_{mn}(Re)$ and the flow field are obtained simultaneously by a parametric physics-informed neural network [21, 22] with inputs $(x, y, \tau, \widetilde{Re})$, where τ is the nondimensional time of Eq. (1) sampled over the training window $[0, 1]$, and a multi-objective loss combining the momentum and continuity residuals, the boundary conditions, a viscous-dissipation objective, the actuation-norm constraint $(E_{\mathrm{act}} - E^*)^2$, a spatial-variance regularizer and a Reynolds-separation term. The identification is performed once over $Re \in [100, 1000]$; the branch representatives at $Re = 500$, $E^* = 0.25$ are then exported on one forcing period, i.e. as the same Fourier series evaluated over $0 \le t_{\mathrm{phys}} \le T = 2$, and transferred without retraining. The network architecture, the sampling protocol, the loss weights, the training schedule and the branch-classification rules are reported in the Supplementary Material (Secs. S1–S2) and in Ref. [38]; the network identifies candidate spatio-temporal wall-actuation profiles, which are subsequently evaluated with the independent MRT-LBM solver of Sec. 4.2, and the present paper uses the identified controls as physical boundary data and does not re-examine the identification procedure. What matters here is only that the exported profile is the one the optimizer produced, without harmonic substitution or spatial idealization; where the norm of the exported waveform differs from the target it is rescaled by a single constant factor before imposition (the largest such factor is 0.7%, for B_1),

Table 1: Actuation modes and hydrodynamic states compared in this study ($Re = 500$, fixed viscosity and geometry). $U_{\max}$ is the maximum of the time-mean lid profile and $Re_{\max} = U_{\max} Re$ the corresponding effective Reynolds number; for the time-dependent controls, the maximum over the whole forcing period is given in parentheses. All stationary profiles and the cycle means of the time-dependent ones carry the same quadratic actuation norm, $\overline{E}_{\mathrm{act}} = 0.25$; the instantaneous norm of B_1, B_2 and B_3 varies over the period. f_2 and f_t are the stationary second-harmonic and total temporal energy fractions of the control coefficients. For the pure first-harmonic control, $f_1 = 1$ and $f_2 = 0$; for the pure second-harmonic control, $f_1 = 0$ and $f_2 = 1$. The two B_1 rows share the same series but not the same denominator, and this is why their f_2 values differ: for the full series f_2 is the stationary second-harmonic coefficient $\mathcal{E}_{20}$ divided by the *total* control energy, which is 92.2%, the value tabulated for the identification campaign in Supplementary Secs. S3.2 and S3.3; renormalising the series onto its own time mean, as the mean-profile row does, removes the temporal share and raises the same coefficient to 93.4%. The parent archive of B_1 is the seedless reference Fourier export of the parent campaign; Supplementary Sec. S4.6 gives its file name, checksum and rescaling factor, since B_1 is identified by that export rather than by a random seed.

Actuation	Structure	$U_{\max}$	$Re_{\max}$	f_2	f_t	Role
Uniform lid	constant, sharp corners	0.50	250	—	0	reference
Uniform lid, $\delta = 0.05$	constant, tapered	0.52	258	—	0	grid-convergent ref.
$k = 1$	$\sin \pi x$	0.71	354	0.00	0	control
$k = 2$	$\sin 2\pi x$	0.71	354	1.00	0	control
B_1 mean profile	mean of B_1	0.78	390	0.93	0	stationary
B_1 full series	Fourier series	0.77 (0.82)	387 (408)	0.92	0.013	branch
B_2 seed 5	temporal	0.04 (1.23)	19 (614)	0.00	1.00	unsteady
B_3 seed 0	mixed	0.62 (1.16)	310 (582)	0.52	0.45	intermediate

and the realized cycle-mean norm is 0.2500 for all three branches.

4.2 Independent MRT-LBM evaluation

The flows are computed with a multiple-relaxation-time D2Q9 lattice-Boltzmann solver [31, 32] that is independent of the identification framework: a Cartesian lattice instead of scattered collocation, a different discretization of the same equations, and a different iteration strategy. Collision is performed in moment space with a diagonal relaxation matrix whose non-viscous rates take the conventional D2Q9 values (Supplementary Sec. S4.1) [35], and the moving lid is imposed on the two wall-facing populations of the lid row by a half-way rebounce whose momentum injection is linear in the wall speed, $f_q = f_{\bar{q}}^{*} + 2 w_q \rho_w (\mathbf{c}_q \cdot \mathbf{u}_w)/c_s^2$ [34] (Supplementary Sec. S4.1 gives the two updates, their place in the collision–streaming sequence, and why the resulting difference between the lid-node and wall-plane velocities is a sampling offset rather than a slip); it is a moving-wall rebound rather than the non-equilibrium bounce-back velocity condition of Zou and He [33]. The solver reproduces the benchmark centerline velocities of Ghia et al. [1] with $L_2(u) = 0.0051$ and $L_\infty(u) = 0.0128$ at $Re = 100$, and $L_2(u) = 0.0141$ and $L_\infty(u) = 0.0300$ at $Re = 1000$ on the 256×256 lattice (Supplementary Sec. S4.2), which fixes the resolution used for the controlled flows. The lattice reference velocity is $U_{\mathrm{ref}} = 0.05$, and the largest instantaneous lid speed imposed in the campaign is $1.23\, U_{\mathrm{ref}}$, so the maximum Mach number is $\mathrm{Ma} \approx 0.11$, within the incompressible limit.

The temporal wall actuation is defined over the physical nondimensional period $T = 2$, with Fourier components $\cos(2n\pi t/T) = \cos(n\pi t)$. The exported control coefficients are evaluated over one complete period and imposed on the independent MRT-LBM solver using a resolution-consistent temporal mapping. With lattice spacing $\Delta x = 1/N$ and lattice reference velocity $U_{\mathrm{ref}} = 0.05$, the nondimensional time increment is $\Delta t = U_{\mathrm{ref}}/N$. Accordingly, the number of lattice time steps per forcing period is

$$N_T = \frac{TN}{U_{\mathrm{ref}}} = 40N. \tag{14}$$

For the $N = 256$ simulations, this corresponds to $N_T = 10\,240$ lattice steps per forcing period,

and $N_T = 20\,480$ at $N = 512$; the temporal campaign is run at $N = 256$ and the three branch captures are replicated at $N = 512$ under this same mapping. This mapping preserves the same physical forcing frequency across spatial resolutions: the fundamental actuation frequency is $\omega_1 = 2\pi/T = \pi$ in cavity units for every lattice, and the oscillatory boundary-layer thickness $\delta_1 = \sqrt{2\nu/\omega_1} = 0.036$, i.e. about nine lattice cells at $N = 256$ and eighteen at $N = 512$, is resolved by the same margin at every resolution. Each of the 256 samples of the exported waveform is held for $N_T/256$ consecutive lattice steps, i.e. 40 at $N = 256$ and 80 at $N = 512$, so its physical duration, $T/256$, is resolution-independent and the staircase it imposes is common to both grids, so the imposed signal is periodic to machine accuracy over the sampling window (Supplementary Sec. S4.4).

Two spacings enter the calculation and are kept distinct. The hydrodynamic spacing is $\Delta x = 1/N$: with the half-way wall condition it is the spacing that fixes Δt, N_T and the lattice viscosity through $\nu_{\text{phys}} = \nu_{\text{latt}}\Delta x^2/\Delta t = 1/Re$, and it is the spacing in terms of which δ_1 is counted in cells above. The cycle-mean integrals are evaluated on the node coordinates of the archived fields, $x_j = y_j = j/(N-1)$, whose spacing is $h = 1/(N-1)$ and which place the outermost nodes on the walls; the two spacings differ by 0.39% at $N = 256$ and 0.20% at $N = 512$. Because the dissipation pairs two finite-difference derivatives with one area element, and the lid power pairs one derivative with one line element, that factor cancels identically in $\overline{D}$, $D(\bar{u})$, D_{fluct} and $\overline{P}_{\text{lid}}$: recomputing the three branch captures with h replaced by $1/N$ changes them by less than 5×10^{-16} in relative terms at both resolutions. It survives in quantities that carry a measure without a paired derivative, where it is a common factor: 0.78% and 0.39% on the kinetic energies, 0.39% and 0.20% on a pointwise derivative such as the steady-streaming forcing, and, for the same reason, 0.39% and 0.20% on a quotient of two terms of different differential order, such as the ratio of that forcing to the mean viscous term (Supplementary Sec. S4.7). Being multiplicative and branch-independent, the factor alters no branch-to-branch ratio, no ranking and no figure reading, and it is an order of magnitude below the grid-refinement variation of Sec. 4.3 (Supplementary Sec. S4.5).

Each actuation is advanced from rest with the lid switched on at $t = 0$. The end of the settling phase is fixed by a drift test on the cycle-averaged dissipation: successive three-period windows are compared until their mean changes by less than 2×10^{-3} in relative terms, which terminates the transient after 24 forced periods for all three branches; 32 consecutive periods are then retained for the statistics. The same protocol is used for every case, so the comparison does not depend on the transient length. Instantaneous dissipation, kinetic energy and lid power are accumulated at every eighth step, and the cycle mean is a direct time average over the 32 sampled periods rather than a phase-quadrature reconstruction: an eight-bin phase quadrature of the same data reproduces the cycle-mean dissipation to within 0.004%, but it under-estimates the peakier lid power by 0.14%, 5.5% and 3.9% for B_1, B_2 and B_3, so the quadrature is used only as a cross-check. The wall shear in Eq. (4) is evaluated with a first-order one-sided derivative of the recovered wall velocity, and the resulting estimator uncertainty is quantified in Sec. 6.5.

4.3 Grid convergence and uncertainty

The three stationary controls and both uniform references are computed on $N = 128$, 256 and 512 lattices. The observed order of convergence p_{obs} and the fine-grid grid-convergence index are obtained by Richardson extrapolation with a safety factor $F_s = 1.25$ [36, 37]. A triple is flagged, and its extrapolated value discarded, when it is non-monotone, when $p_{\text{obs}} \notin [0.1, 4]$, or when the GCI exceeds 100%: in those cases the smooth-error assumption underlying the estimator is not credible. The estimator is applied both to the absolute dissipation and to the dissipation ratios, since the ratios are the quantities used in the physical discussion and their errors need not cancel. The time-dependent comparison is performed with one realization per branch on the $N = 256$ lattice and replicated on $N = 512$ under the same temporal protocol; each branch representative is the periodic response to the single exported waveform of Sec. 3.2, and the quoted

Table 2: Stationary controls at $Re = 500$ and matched quadratic actuation norm $E^* = 0.25$ on three grids. ε_N is the integrated viscous dissipation; p_{obs} and GCI_{512} are the observed order and the fine-grid grid-convergence index of the absolute dissipation ($F_s = 1.25$). The last column gives the dissipation normalized by the sharp and by the corner-regularized uniform lid on the same grid (the two values are separated by a slash). The B_1 mean profile is used as computed ($E_r = 0.2503$). A dagger marks a flagged triple, for which the extrapolated value is not used. $U_{\max}$ and $Re_{\max}$ are repeated here because the norm constraint fixes the rms wall velocity (0.5) but not the peak.

Control	$U_{\max}$	$Re_{\max}$	ε_{128}	ε_{256}	ε_{512}	p_{obs}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\mathrm{ref}}$ (sharp / reg.)
Uniform lid (sharp)	0.50	250	0.00774	0.00948	0.01102	0.18	130.71†	1.000 / 1.272
Uniform lid, $\delta = 0.05$	0.52	258	0.00717	0.00815	0.00866	0.96	7.79	0.786 / 1.000
$k = 1$ ($\sin \pi x$)	0.71	354	0.00635	0.00688	0.00713	1.09	3.85	0.647 / 0.823
$k = 2$ ($\sin 2\pi x$)	0.71	354	0.01012	0.01129	0.01185	1.04	5.62	1.076 / 1.368
B_1 mean profile	0.78	390	0.01067	0.01189	0.01250	1.01	6.01	1.134 / 1.443

numbers are therefore properties of those controls rather than of the identification classes they represent; that restriction is stated wherever those numbers are used. The replication is reported as a *grid-refinement sensitivity*: two grids bound the variation of the cycle means, they do not determine a convergence order, so no extrapolation and no convergence index is computed for them. The refined runs raise $\overline{D}$ by 5.4%, 7.3% and 7.1% for B_1, B_2 and B_3, preserve the ordering $B_2 > B_1 > B_3$, and halve the power-balance residual of Sec. 6.5; it is tabulated in Sec. S4.5. The replication supplies scalar diagnostics only: every field in this paper, and every phase-resolved quantity, is an $N = 256$ capture. Within a run, by contrast, the statistics are not single-sample: the cycle means of Sec. 6.2 average 32 forced periods and their block uncertainty is reported in Sec. 6.5.

5 Energetic response of stationary wall-actuation modes

5.1 Comparison at matched actuation norm

Table 2 and Fig. 3 give the integrated dissipation of the five stationary controls at $Re = 500$ and $E^* = 0.25$. On the finest grid the first harmonic is the only control that lowers the dissipation relative to the classical uniform lid: $\varepsilon_{512} = 0.00713$ against 0.01102, a ratio of 0.647. The second harmonic raises it to 0.01185 (ratio 1.076) and the B_1 mean profile to 0.01250 (ratio 1.134). The ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is the central stationary result, and it is the reverse of the fixed-lid-velocity ordering reported in Ref. [38], where both harmonic profiles were less dissipative than the uniform lid and the second harmonic was the least dissipative of the three. The reversal is not a refinement of that result but a consequence of the constraint: at equal lid speed the harmonics have half the norm of the uniform lid, and restoring the norm raises their wall velocity by $\sqrt{2}$ and their effective Reynolds number with it.

The ratios built on the sharp uniform lid are not converged estimates of the shape effect, because the reference itself does not converge: its increments between successive grids decay by only 12% per refinement ($p_{\mathrm{obs}} = 0.18$, GCI 131%), and the formal Richardson limit, 2.25×10^{-2}, is more than twice the finest-grid value and is not a credible continuous-limit dissipation. The three other controls converge, with $p_{\mathrm{obs}} = 1.09$, 1.04 and 1.01 and fine-grid GCIs of 3.9%, 5.6% and 6.0%. Normalizing by the corner-regularized lid instead gives ratios of 0.823, 1.368 and 1.443 at $N = 512$, i.e. -17.7%, $+36.8\%$ and $+44.3\%$, with Richardson estimates 0.802, 1.347 and 1.427 and ratio GCIs of 3.1%, 1.9% and 1.4%. The sign of the ordering is the same under both references and at all three resolutions; only its magnitude changes, because the non-converged corner contribution inflates the denominator of the sharp-lid ratios. We therefore quote the regularized ratios as the better-defined estimate of the shape effect and the sharp-lid ratios as the values directly comparable with the classical uniform-lid baseline, and we do not use the sharp-lid ratio for $k = 1$ as a quantitative reduction: -35.3% overstates the -17.7% obtained against a grid-convergent

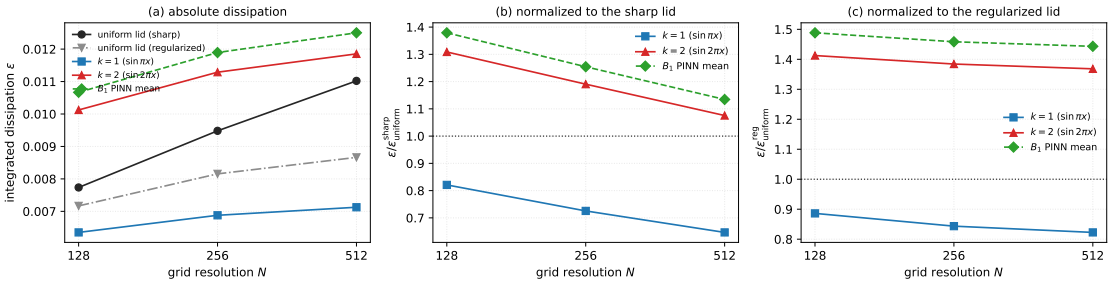


Figure 3: Grid convergence of the stationary comparison at $Re = 500$ and matched actuation norm $E^* = 0.25$. (a) Integrated dissipation versus resolution for the four stationary controls and for the corner-regularized uniform lid. (b) Dissipation normalized by the sharp uniform lid on the same grid: this reference does not converge, and the ratios drift by 18–21% over the triple. (c) Dissipation normalized by the regularized lid: the three ratio triples are monotone and converge with GCIs of 1.4–3.1%. The sign of the ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is observed at every resolution in both (b) and (c); its magnitude is reference-dependent.

reference. The separation between $k = 2$ and the B_1 mean profile (1.368 against 1.443, about 5%) exceeds their GCIs but not by a wide margin, so they are reported as approximately equal rather than strictly ordered.

5.2 The corner singularity and the reference dependence

The non-convergence of the uniform lid is a property of its corners, not of the uniform profile as such. Its dissipation grows by 16% between $N = 256$ and $N = 512$, the signature of a boundary-layer singularity that the grid cannot resolve. Replacing the discontinuity by the cosine taper of Eq. (11) removes 21% of the dissipation at $N = 512$ and, more importantly, changes the growth between the two finest grids from 16% to 6%, i.e. the expected first-order rate: $p_{\mathrm{obs}} = 0.96$, GCI 7.8%, extrapolated $\varepsilon_\infty = 9.20 \times 10^{-3}$. A continuous lid therefore converges, and the singular one does not. This has two consequences for the interpretation. First, a comparison of actuation shapes that uses the discontinuous lid as denominator mixes the shape effect with a resolution artefact; the regularized reference is the only one that separates them. Second, the corner is an energetic feature in its own right: the uniform lid concentrates a large part of its dissipation in two small regions whose contribution grows with resolution, so any ranking that depends on the corner peak is resolution-dependent by construction.

The practical conclusion is that two numbers must be reported for each profile, not one. The sharp-lid ratio answers the question "what does this actuation do relative to the classical cavity?" and is the quantity a reader will compare with the literature. The regularized ratio answers "what is the effect of the spatial shape at fixed actuation norm?" and is the quantity that supports a physical statement about shape. Both are given in Table 2, and the text uses the second for every claim about ordering or magnitude.

5.3 Local mechanism of the second-harmonic penalty

The ranking has a local origin that can be read directly from the fields (Fig. 4). The uniform lid concentrates its dissipation in the two upper corners, where the imposed tangential velocity jumps to zero: the peak strain-rate magnitude there is 169 in units of U_0/L, against 15.0 for $k = 1$ and 23.4 for $k = 2$, and those corner layers are the brightest features of the dissipation field. Both harmonics remove the jump, and their dissipation is spread over the lid shear layer and the side walls instead. But the second harmonic pays for this in the interior. Its lid generates a pair of counter-rotating cells of nearly equal strength, separated by a wall-to-wall separatrix at $x = 0.5$, each enclosing its own low-strain core; the uniform and first-harmonic lids recover the classical single primary vortex with a corner eddy. The reorganization shifts the strain-rate

distribution towards larger values: the interior mean $\langle|\mathbf{S}|\rangle$ is 1.450 for $k = 2$, against 1.039 for the uniform lid and 1.088 for $k = 1$, a 40% increase at identical actuation norm. We note that $\langle|\mathbf{S}|\rangle$ is a first-moment diagnostic, whereas the dissipation is proportional to $|\mathbf{S}|^2$; a full spatial decomposition of D by zones (corners, wall layers, interior) would provide a more quantitative mechanism argument. The corner peak of 169 at $N = 512$ is a grid-dependent quantity due to the velocity discontinuity singularity; its exact value should not be over-interpreted, though its presence and the effect of regularization are physically meaningful.

The trade-off is explicit. The first harmonic removes almost all of the corner peak (169 to 15.0) at a cost of 4.7% in interior mean strain, and is a net reduction. The second harmonic removes slightly less of it (23.4) while raising the interior mean by 39.6%, and is a net penalty: it buys a symmetric two-cell state with a stronger interior strain field, and the corner singularity it removes is not costly enough to compensate. Relative to the uniform lid, the increase in interior mean strain is only 4.7% for $k = 1$, but reaches 39.6% for $k = 2$. The latter increase is approximately 8.4 times larger, even though the second harmonic does not reduce the peak strain below that of the first harmonic (23.4 against 15.0). Additional identification-sensitivity and control-representation studies, including loss-term ablation, network-capacity sensitivity, control-space truncation, aspect-ratio dependence and Fourier versus modified-Chebyshev parametrizations, are documented in Supplementary Sec. S3 and Tables S4–S9. These studies characterize the robustness of branch identification; the dissipation ordering discussed here is established independently by the MRT-LBM calculations and their grid and estimator checks in Supplementary Secs. S4.4–S4.8.

6 Cycle-mean dissipation of time-dependent actuation branches

6.1 Why the dissipation of the mean flow is not the cycle mean

For the stationary controls of Sec. 5 the distinction between $D(\overline{\mathbf{u}})$ and $\overline{D}$ is empty, because the response is steady. For the time-dependent branches it becomes the central issue. Equation (8) is exact: the cycle mean exceeds the mean-field value by the dissipation of the fluctuations, and there is no reason for that term to be small when the actuation reverses sign. A mean-field diagnostic answers the question "how much would this flow dissipate if it were steady?"; it does not answer "how much does it dissipate?". Figure 5 and Table 3 show how far apart the two answers can be; supplementary Tables S13 and S14 provide additional cycle-averaged statistics and iso-energetic comparisons, while supplementary Figs. S1 and S2 provide the phase-resolved field captures.

6.2 Cycle-mean comparison of the three branches

Table 3 collects the result. The four stationary controls span less than a factor of two, from 6.88×10^{-3} for $k = 1$ to 1.19×10^{-2} for the B_1 mean profile, and their mean-field and cycle-mean values coincide by construction; for the B_1 full series the two are 1.172×10^{-2} and 1.193×10^{-2}, a difference of 1.8%, which is the residual effect of its small temporal content ($f_t = 0.013$). The branches with substantial temporal content separate through their fluctuating part, but not through their total. B_2 dissipates 1.441×10^{-2} over the cycle while its mean field dissipates 1.016×10^{-3}, a ratio of 14.2; its mean field holds only 25% of the cycle-mean kinetic energy, so the mean motion is a shallow recirculation under the lid, not a weakened copy of the instantaneous flow. B_3 is intermediate in structure: $D(\overline{\mathbf{u}}) = 5.41 \times 10^{-3}$ against $\overline{D} = 1.169 \times 10^{-2}$, a ratio of 2.16, with a fluctuating part that carries 53.7% of its total cost. On the cycle average, and at the same actuation norm, the three branches lie within a factor 1.23 of one another: B_2 costs 20.7% more than B_1 and 23.2% more than B_3, while B_1 and B_3 are nearly equal, 2.1% apart at $N = 256$ and 0.5% apart at $N = 512$, although their temporal fractions span 0.013, 0.448 and 0.998. The most oscillatory branch remains 2.1 times as dissipative as the least dissipative stationary control $k = 1$, so the unsteady actuations do not fall below the stationary band; they cluster at its upper edge. Each quoted value is a single realization of one exported waveform (Sec. 4.3). Repeating

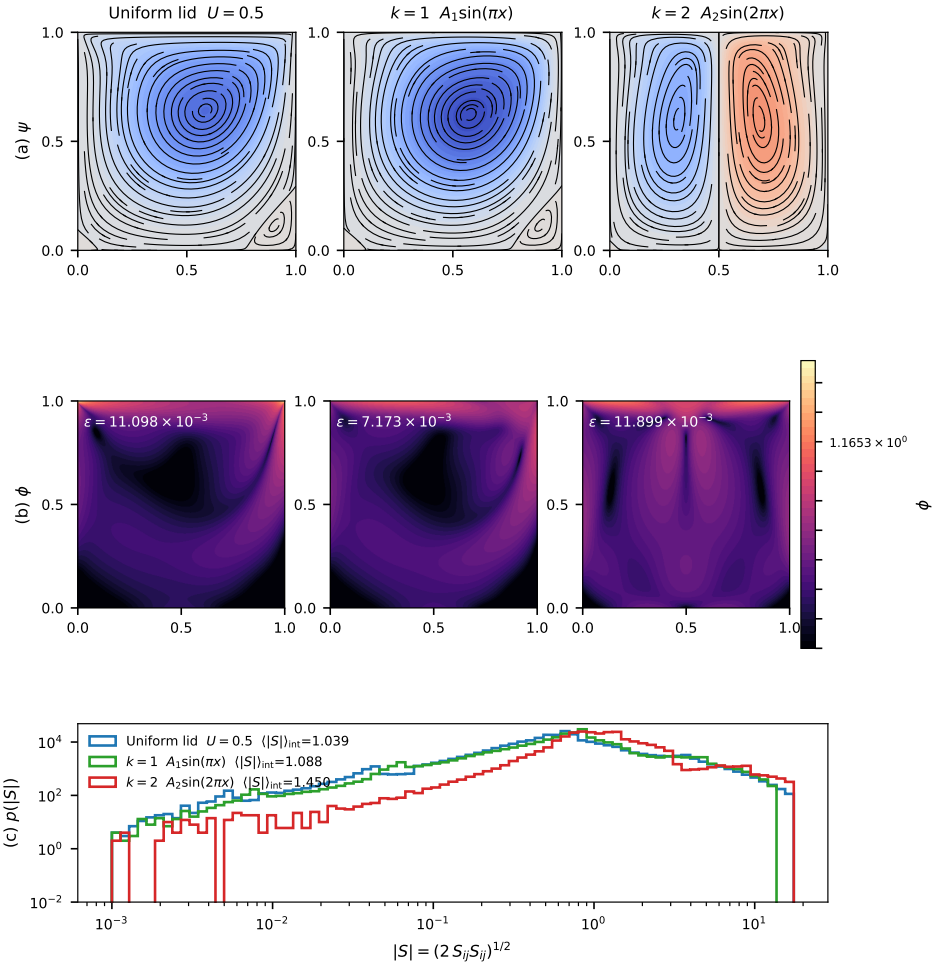


Figure 4: Local origin of the stationary ranking at $Re = 500$, $N = 512$ and matched actuation norm, on scales common to the three columns. (a) Stream function of the steady response with the streamline topology. (b) Local viscous dissipation $\phi = \nu(2S_{11}^2 + 2S_{22}^2 + 4S_{12}^2)$ on a logarithmic scale spanning six decades below the global maximum; the annotation gives the domain integral. (c) Probability density of the strain-rate magnitude over the cavity interior (three boundary cells excluded), with the interior mean $\langle|\mathbf{S}|\rangle_{\rm int}$. The uniform lid peaks at the corners ($|\mathbf{S}|_{\max} = 169$); $k = 1$ removes the peak at almost no interior cost ($\langle|\mathbf{S}|\rangle_{\rm int} = 1.088$ against 1.039); $k = 2$ removes less of it (23.4) and raises the interior mean to 1.450, which is the origin of its dissipation penalty.

the three captures on the $N = 512$ lattice under the same protocol raises all three cycle means by 5.4–7.3%, leaves the ordering unchanged, and narrows the B_1–B_3 separation from 2.1% to 0.5%: the near-equality of those two branches, which their temporal fractions of 0.013 and 0.448 alone would not suggest, is strengthened rather than weakened by the refinement (Sec. S4.5).

The corner-regularized uniform lid is used as the grid-convergent reference for the stationary grid study (Sec. 5) and is therefore not included in the seven-control comparison above, which retains the sharp uniform lid as the canonical cavity reference.

Three statements follow, and they are the ones the paper retains. First, the mean-field functional is not a surrogate for the cycle mean: for the actuations whose wall velocity reverses in time it is too small by a factor 2.2 (B_3) to a factor 14 (B_2), and the difference is exactly the dissipation of the fluctuations, which carries 53.7% and 92.9% of the respective totals. Two branches whose totals differ by only about 2.1% can rest on opposite compositions of that total. Second, the temporal fraction f_t of the actuation does not predict the cycle-mean cost monotonically: B_2 ($f_t = 0.998$) is the most dissipative of the three, while B_1 ($f_t = 0.013$) and

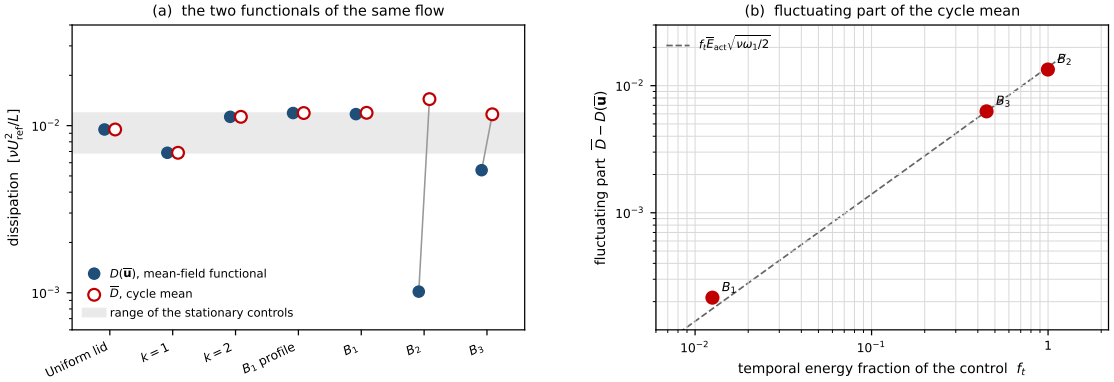


Figure 5: Iso-energetic comparison of the seven controls at $Re = 500$, $\overline{E}_{\mathrm{act}} = 0.25$ and $N = 256$. (a) The two dissipation functionals of each control: the dissipation of the time-mean field (filled markers) and the cycle mean (open markers), joined by the gap between them; the two coincide for the stationary controls, whose span is shaded. (b) That gap—the fluctuating part of the cycle mean—against the temporal energy fraction f_t of the lid control, together with the first-harmonic Stokes estimate $f_t\overline{E}_{\mathrm{act}}\sqrt{\nu\omega_1/2}$ (dashed line, no adjustable constant). The cycle-mean totals of the three branches span a factor of at most 1.23 and lie inside the stationary band except for B_2, while the fluctuating parts they conceal span two orders of magnitude and follow the boundary-layer estimate. The branch values are single realizations.

Table 3: Dissipation of the time-mean field and cycle-mean dissipation at $Re = 500$, $N = 256$ and cycle-mean actuation norm $\overline{E}_{\mathrm{act}} = 0.25$. $D(\overline{\mathbf{u}})$ is the functional evaluated on the time-mean field; $\overline{D}$ is the average of the instantaneous dissipation over the 32 sampled periods of the periodic response and is the performance index relevant to a periodic flow. The two coincide only in the steady limit. A_ε is the half-spread of the phase-resolved $D(t)$ over the cycle, i.e. the variability concealed by a single cycle-mean number. The B_1 mean profile row is stationary by construction and is used as computed, with $E_{\mathrm{act}} = 0.2503$; the B_1 full row is the identified series renormalised to $\overline{E}_{\mathrm{act}} = 0.25$, whose own time mean therefore carries 0.2468, so those two rows are a comparison at nearby but not identical mean forcing norms rather than a strictly iso-energetic one (Sec. S4.3). The three branch rows are single realizations.

Control	f_t	$\overline{E}_{\mathrm{act}}$	$D(\overline{\mathbf{u}})$	$\overline{D}$	$\overline{D}/D(\overline{\mathbf{u}})$	A_ε
Uniform lid	0.000	0.250	9.48×10^{-3}	9.48×10^{-3}	1.00	0
$k = 1$ ($\sin \pi x$)	0.000	0.250	6.88×10^{-3}	6.88×10^{-3}	1.00	0
$k = 2$ ($\sin 2\pi x$)	0.000	0.250	1.13×10^{-2}	1.13×10^{-2}	1.00	0
B_1 mean profile	0.000	0.250	1.19×10^{-2}	1.19×10^{-2}	1.00	0
B_1 full series	0.013	0.2500	1.172×10^{-2}	1.193×10^{-2}	1.02	1.8×10^{-3}
B_2 seed 5	0.998	0.2500	1.016×10^{-3}	1.441×10^{-2}	14.2	1.0×10^{-2}
B_3 seed 0	0.448	0.2500	5.413×10^{-3}	1.169×10^{-2}	2.16	4.3×10^{-3}

B_3 ($f_t = 0.448$), whose oscillatory actuation energies differ by a factor of 36 (3.1×10^{-3} against 0.1120 at the same $E^* = 0.25$), cost the same to within 2.1%. Third, the ordering of the totals ($B_2 > B_1 \simeq B_3$) is not the ordering of the mean fields, which reads $B_1 > B_3 > B_2$: the branch that suppresses its mean flow most strongly is the one whose total cost stays essentially equal to those of its structurally different companions, because what it loses in mean-field dissipation it regains, term for term, in fluctuating dissipation (Eq. (8)).

6.3 Order-of-magnitude scaling of the temporal cost

The cycle-mean cost of the unsteady branches is not an accident of the identification; it is already visible in the boundary-layer scaling of the actuation. For a lid harmonic of peak amplitude A_n

and angular frequency ω_n, the second problem of Stokes gives a time-mean dissipation per unit lid area $\frac{1}{2}A_n^2\sqrt{\nu\omega_n/2}$, with oscillatory layer thickness $\delta_n = \sqrt{2\nu/\omega_n}$. Summing over the temporal harmonics of the actually imposed lid series,

$$\overline{D}_{\mathrm{St}} = \sum_{n\geq 1} E_n \sqrt{\frac{\nu\omega_n}{2}}, \qquad E_n = \frac{1}{2}\int_0^1 A_n^2(x)\,dx, \tag{15}$$

where $A_n(x)$ is the amplitude of temporal harmonic n of the lid signal and E_n is the actuation energy it carries, so that $\sum_{n\geq 1} E_n = f_t \overline{E}_{\mathrm{act}}$. The decomposition of the three imposed lid signals reproduces the realized norm of Table 3 (0.2500 for each branch) and its temporal fractions to within 0.001 (0.013, 0.998 and 0.448), so it can also be read as an independent check of those numbers. At $Re = 500$ ($\nu = 2\times10^{-3}$) and $T = 2$ ($\omega_n = n\pi$) Eq. (15) gives $\overline{D}_{\mathrm{St}} = 1.41\times10^{-2}$ for B_2, 6.38×10^{-3} for B_3 and 1.81×10^{-4} for B_1, against measured cycle means of 1.44×10^{-2}, 1.17×10^{-2} and 1.19×10^{-2}.

Three points follow. First, the boundary-layer estimate should be compared not to the total cycle-mean dissipation but to its fluctuating component $D_{\mathrm{fluct}} = \overline{D} - D(\overline{\mathbf{u}})$, which the decomposition of Sec. 2.3 isolates exactly. The measured fluctuating parts are 1.339×10^{-2} for B_2 and 6.28×10^{-3} for B_3, against Stokes estimates of 1.41×10^{-2} and 6.38×10^{-3}: the estimate reproduces the fluctuation cost of the lid-oscillatory and mixed branches (B_2 and B_3, whose lid reverses over the whole span and over the central 36% of it; Sec. 3.2) to within approximately 5% and 2%, respectively, with no adjustable constant. It is a half-space model of the oscillatory wall layer, so this agreement speaks to the origin of the fluctuating cost rather than to a spatial decomposition of the measured dissipation. For B_1 the estimate is 1.81×10^{-4} against a measured fluctuation of 2.15×10^{-4}, i.e. it falls 16% short of a quantity that is itself a small difference of two larger ones and should not be over-read. Second, the scaling explains the headline ranking without invoking the field solution. The three controls concentrate essentially all of their oscillatory energy on the first temporal harmonic (98% for B_2, 96% for B_3, 93% for B_1), so Eq. (15) reduces to $\overline{D}_{\mathrm{St}} \simeq f_t \overline{E}_{\mathrm{act}}\sqrt{\nu\omega_1/2}$: the fluctuating cost is set by the oscillatory actuation energy alone, not by how mixed the spatial structure is. Adding this term to the mean-field dissipation of the same run gives 1.190×10^{-2}, 1.51×10^{-2} and 1.180×10^{-2} for B_1, B_2 and B_3, against measured totals of 1.193×10^{-2}, 1.441×10^{-2} and 1.169×10^{-2}, i.e. within 0.3%, 5% and 0.9%. The near-degeneracy of the three costs is then a balance sheet rather than a coincidence: B_1 spends 1.17×10^{-2} in its mean field and almost nothing in fluctuations, B_3 spends 5.4×10^{-3} and 6.3×10^{-3}, and B_2 spends 1.0×10^{-3} and 13.4×10^{-3} — three different compositions, one total, except for B_2, which escapes the stationary floor and lands 21% above it. A direct test of the $\sqrt{\omega_n}$ dependence is left for future work: it requires single-harmonic excitations at $n = 1, 2, 4$ with identical oscillatory energy, whereas Supplementary Sec. S4.5 only tabulates the harmonic budgets of the three controls already used here. Third, the estimate also makes the resolution requirement quantitative: at $Re = 500$ the fundamental layer is $\delta_1 = 0.0357L$, i.e. about 9.1 cells at $N = 256$, and $\delta_4 = 0.0178L$, about 4.6 cells, so the first-harmonic-dominated budgets above are resolved by construction, while harmonics above $n \approx 4$ are marginal at this resolution.

6.4 Flow organization behind the cycle-mean cost

The field-level organization explains the numbers (Figs. 6 and 7). The B_1 time-mean flow is the familiar pair of counter-rotating cells centred at $(0.31, 0.53)$ and $(0.69, 0.57)$, with $\langle\psi\rangle \in [-0.0363, 0.0355]$ and $\max|\langle\mathbf{u}\rangle| = 0.734$; half of its mean kinetic energy lies below $y = 0.671$ and only 30% below $y = 1/2$, so the cells fill the cavity. Its instantaneous states depart from that mean by at most 10% in field norm and its kinetic-energy fluctuation ratio is $R_K = 1.6\times10^{-3}$, so it is quasi-stationary in the operational sense. The B_2 mean field is 3.1 times weaker in stream function ($\max|\langle\psi\rangle| = 0.0116$) and five times weaker in peak speed (0.142, or 19% of the B_1 value), and it is confined: its mean extrema sit at $y = 0.85$ and $y = 0.86$, and half of the mean kinetic energy lies above $y = 0.893$. Its streamwise fluctuation, by contrast, reaches 0.728—5.1 times its

own mean peak and 7.7 times the B_1 fluctuation level, of the same order as the B_1 mean peak itself. The B_3 flow keeps an intermediate mean circulation ($\max|\langle\mathbf{u}\rangle| = 0.598$, 81% of B_1) but places it close to the lid, with extrema at $(0.20, 0.86)$ and $(0.77, 0.80)$ and half of the mean kinetic energy above $y = 0.888$; its fluctuation reaches 0.532. The mean field holds 99.8%, 25% and 87.5% of the cycle-mean kinetic energy for the three branches, respectively.

"Weak mean flow" is therefore not "weak flow". The controls with a strong temporal component do not produce a small perturbation around a mean recirculation: they produce responses whose fluctuations dominate the instantaneous field while the mean is nearly zero. A figure that normalized each panel by its own maximum, or a diagnostic that used only the mean field, would hide exactly the contrast that defines the two unsteady branches.

That contrast has a dynamical reading. For a time-periodic field the mean flow is not driven directly by the lid but by the divergence of the fluctuation Reynolds stress, $\mathbf{F} = -\nabla\cdot\langle\mathbf{u}'\otimes\mathbf{u}'\rangle$, and the steady mean momentum equation reads $\nu\nabla^2\langle\mathbf{u}\rangle = \nabla\langle p\rangle - \mathbf{F} + \mathbf{c}_{\mathrm{adv}}$ with $\mathbf{c}_{\mathrm{adv}} = (\langle\mathbf{u}\rangle\cdot\nabla)\langle\mathbf{u}\rangle$; its curl reads $\nu\nabla^2\langle\omega\rangle = -\nabla\times\mathbf{F} + \nabla\times\mathbf{c}_{\mathrm{adv}}$, and in the Stokes-order reading, with the mean advection dropped, $\nu\nabla^2\langle\omega\rangle = -\nabla\times\mathbf{F}$. Equivalently, the evolutionary form is $\partial_t\langle\omega\rangle = \nu\nabla^2\langle\omega\rangle + S_\omega - \nabla\times\mathbf{c}_{\mathrm{adv}}$ with $S_\omega \equiv \nabla\times\mathbf{F} = \partial_x F_y - \partial_y F_x$, and the field mapped below is that source S_ω: its extrema are values of $\nabla\times\mathbf{F}$, which at steady state balances the viscous and advective curls with the opposite sign. This is the classical steady-streaming problem [10]. Evaluated on the three captures from eight phase bins, the streamwise normal stress at the lid plane is 0.203 for B_2 and 7.0×10^{-2} for B_3 against 2.2×10^{-3} for B_1, and the resulting forcing has root-mean-square magnitude 5.3×10^{-2} for B_2, i.e. 26% of the root-mean-square of the viscous term $\nu\nabla^2\langle\mathbf{u}\rangle$ of its own mean field (24% at their respective maxima). For B_3 the same ratio is 6.7% (10% at the maxima) and for B_1 it is 0.27%. The source S_ω of B_2 peaks at $+23.7$ just below the lid, at $(0.36, 0.99)$, and that of B_3 at -22.1 at $(0.64, 0.99)$, while the B_1 source stays below 1.1 everywhere; the two unsteady sources have opposite sign and sit on either side of the midline, at heights above the mean extrema reported above, which is consistent with mean cells that hug the lid instead of filling the cavity. The source changes sign across the upper quarter of the domain (47% and 52% of the cells positive for B_2 and B_3), so it feeds a counter-rotating pair rather than a one-signed jet, which is why the B_2 mean is a weak pair of cells and not a unidirectional streaming channel. Two limits bound this reading: the mean pressure gradient and mean advection were not recovered, so the comparison is a magnitude budget of the forcing against the viscous term and not a closed balance; and the eight-bin quadrature lets the self-product of the higher lid harmonics alias into the phase mean, an energy share of 1.2×10^{-4} in B_2, so aliasing cannot account for the factor 92 between the lid normal stresses of B_2 and B_1, nor for the factor 32 between B_3 and B_1. What the diagnostic does establish is structural: the B_2 mean circulation is consistent with a Reynolds-stress-driven steady-streaming mechanism, not a stationary recirculation driven by a steady lid. The spatial distribution of the fluctuation forcing is compatible with the observed near-lid mean cells, although a closed mean-vorticity budget is not yet available. It is therefore legitimate to treat its mean field as energetically negligible while its fluctuations carry nearly the whole cost. The forcing and stress magnitudes are tabulated with their definitions and their limits in Supplementary Sec. S4.7.

6.5 Mechanical energy budget of the branches

Table 4 and Fig. 8 close the comparison with the budget of Eq. (9). The cycle-mean kinetic energy spans a range of a factor 4.8 across the three branches (1.161×10^{-2}, 2.423×10^{-3} and 5.526×10^{-3}), while the cycle-mean dissipation orders them nearly flat (1.193×10^{-2}, 1.441×10^{-2} and 1.169×10^{-2}). Within the period, $D(t)$ varies by a factor 1.43, 7.42 and 2.42 respectively, and the lid power of B_2 is negative for 17% of the period: the flow returns energy through the lid while the Stokes layer spins down. The residual $\overline{P}^{\mathrm{rec}}_{\mathrm{lid}} - \overline{D}$, computed with the wall velocity recovered at the outermost fluid row rather than the prescribed profile (Sec. S4.8) — i.e. across the lid node row, which the half-way boundary condition places half a lattice spacing below the hydrodynamic wall — is -2.8%, -0.6% and -0.4%.

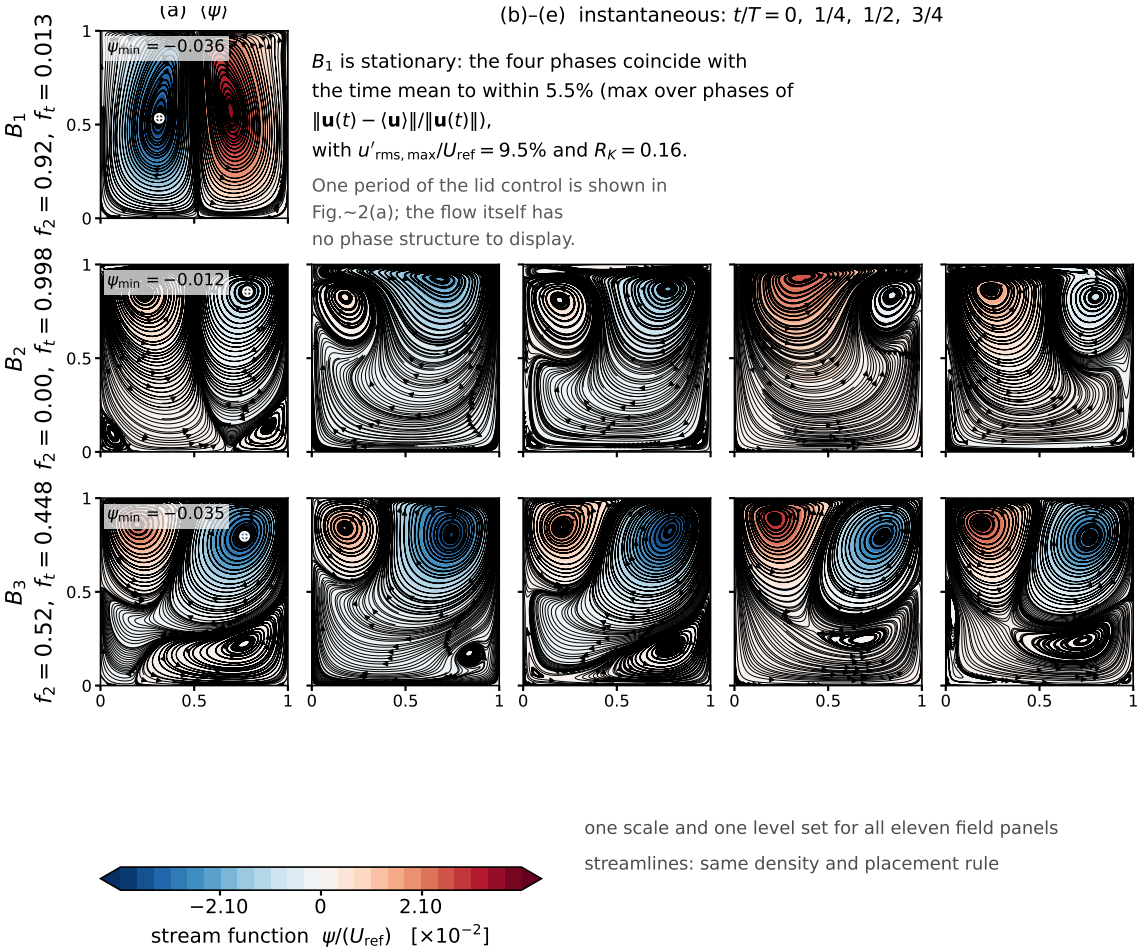


Figure 6: Flow topology of the three branch controls at $Re = 500$, $E^* = 0.25$ and $N = 256$, on one stream-function scale and one set of contour levels common to all panels. Rows: B_1 (full series, $f_t = 0.013$), B_2 (seed 5, $f_t = 0.998$) and B_3 (seed 0, $f_2 = 0.520$, $f_t = 0.448$). (a) Time mean $\langle\psi\rangle$ with the streamlines of the mean flow; the circle marks its negative extremum. (b)–(e) Instantaneous ψ and streamlines at $t/T = 0, 1/4, 1/2, 3/4$ for B_2 and B_3; the four phases belong to the same periodic state, whose phase averages, recomputed over the first and over the second half of the capture, differ by at most 1.5×10^{-5} for B_2 and 2.6×10^{-4} for B_3 in relative field norm. For B_1 the instantaneous states lie within 10% of the mean, so the four panels are replaced by that statement. The common scale is deliberate: the B_2 mean cell is three times weaker in stream function than the B_1 one and is confined to the top decile of the cavity, which is the point.

The residual is two to three orders of magnitude larger than the drift of the sampling window itself: the difference between the kinetic energy averaged over the first and over the last half of the 32 retained periods is at most $5.6 \times 10^{-3}\%$ of $\overline{D}$, and the corresponding drift of $\overline{D}$ is below $1.5 \times 10^{-3}\%$; the residual is therefore not a failure of periodicity. It is the truncation error of the one-sided wall-shear estimate. Two checks bound that error. First, the estimator accumulated at every lattice step and the same estimator evaluated on the eight phase-resolved fields agree to 0.14% for B_1, whose lid power is nearly sinusoidal; for B_2 and B_3 the phase reconstruction is 5.5% and 3.9% below the step-by-step accumulator, the quadrature shortfall already documented in Sec. 4.2, so the table quotes the accumulator. Second, replacing the first-order one-sided derivative by a second-order one on the B_1 mean field moves its residual from -3.06% to $+0.39\%$, i.e. it crosses zero, which shows that the estimator uncertainty is of the order of the residual itself (a few per cent). The same substitution amplifies lattice-scale noise in the thin Stokes layer of the two unsteady branches and is not used. Third, refining the lattice from $N = 256$ to $N = 512$ on the same temporal protocol divides the three residuals by 2.1, 1.9 and 2.4 — the factor of two

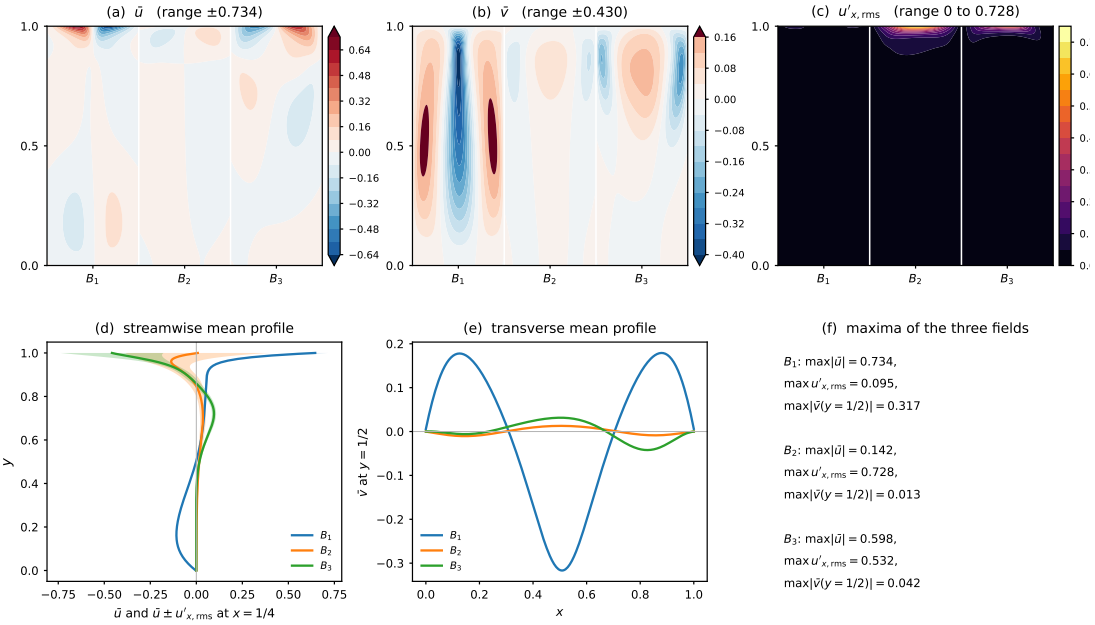


Figure 7: Velocity fields and centreline profiles of the same three captures. Rows: (a) $\bar{u}$, (b) $\bar{v}$ and (c) $u'_{x,\mathrm{rms}}$, each on one colour range for the three columns and a single normalization, with the occupied range written in every panel so that the common scale can be checked. Grey lines are the zero contours of the mean fields. (d) $\bar{u}$ along $x = 1/4$ with the $\pm u'_{\mathrm{rms}}$ band; the midline $x = 1/2$ is not used because the antisymmetric controls have $u \equiv 0$ there. (e) $\bar{v}$ along $y = 1/2$, with $\bar{v}$ on the abscissa. The bands are temporal standard deviations over the thirty-two sampled periods, not confidence intervals.

Table 4: Mechanical energy budget of the three branch controls at $Re = 500$ and $N = 256$, at the common cycle-mean actuation norm $E^* = 0.25$: B_1 is the full series ($f_t = 0.013$), B_2 is seed 5 ($f_t = 0.998$) and B_3 is seed 0 ($f_2 = 0.520$, $f_t = 0.448$). All quantities are cycle averages over the 32 forced periods retained after the settling phase, taken from the instantaneous fields recorded every eighth lattice step; $\overline{D}$ is never replaced by $D(\overline{\mathbf{u}})$. $\sigma(\overline{D})$ is the block standard error over four blocks of eight cycles, expanded by $t_{3,0.975} = 3.182$; it is below 0.1% for all three branches because the response is periodic (the phase averages of the two half-windows differ, in relative L_2 norm of the streamwise field, by 1.9×10^{-3} for B_1, 2.6×10^{-4} for B_3 and 1.5×10^{-5} for B_2, Supplementary Sec. S4.6; and the cycle-to-cycle standard deviation of $\overline{D}$ is 7.0×10^{-6} in relative terms for B_1, 2.4×10^{-8} for B_2 and 1.2×10^{-5} for B_3), so the estimator error of the last column, not the sampling error, is what limits the closure. The power $\overline{P}_{\mathrm{lid}}$ is the numerical estimator $\overline{P}^{\mathrm{rec}}_{\mathrm{lid}}$ computed with the wall velocity recovered at the outermost fluid row, the lid node row $i = N - 1$ [Eq. (5)], not the prescribed velocity; it evaluates the power crossing the *lid node row*, half a lattice spacing below the physical wall, rather than the wall itself, so it provides an indirect verification of the discrete energy budget. The resulting residual $\overline{P}^{\mathrm{rec}}_{\mathrm{lid}} - \overline{D}$ is the first-order truncation of the wall-shear estimator, not a physical imbalance (Sec. 6.5).

Branch	$\overline{E}_{\mathrm{act}}$	$\overline{K}$	$\overline{P}^{\mathrm{rec}}_{\mathrm{lid}}$	$\overline{D}$	$\sigma(\overline{D})$
B_1 (full series)	0.2500	1.161×10^{-2}	1.160×10^{-2}	1.193×10^{-2}	$< 0.1\%$
B_2 (seed 5)	0.2500	2.423×10^{-3}	1.432×10^{-2}	1.441×10^{-2}	$< 0.1\%$
B_3 (seed 0)	0.2500	5.526×10^{-3}	1.164×10^{-2}	1.169×10^{-2}	$< 0.1\%$

that a first-order stencil predicts under $\Delta x \to \Delta x/2$ — and moves them to -1.4%, -0.3% and -0.2%, while the largest near-wall nodal velocity offset drops by 38–47% as the sampling node approaches the wall (Secs. S4.5 and S4.8). The residual is thus *measured* to be a truncation error of the estimator, not merely argued to be one. We therefore retain the first-order estimator for all three branches, report the residual as measured, and present Eq. (10) as an energetic verification

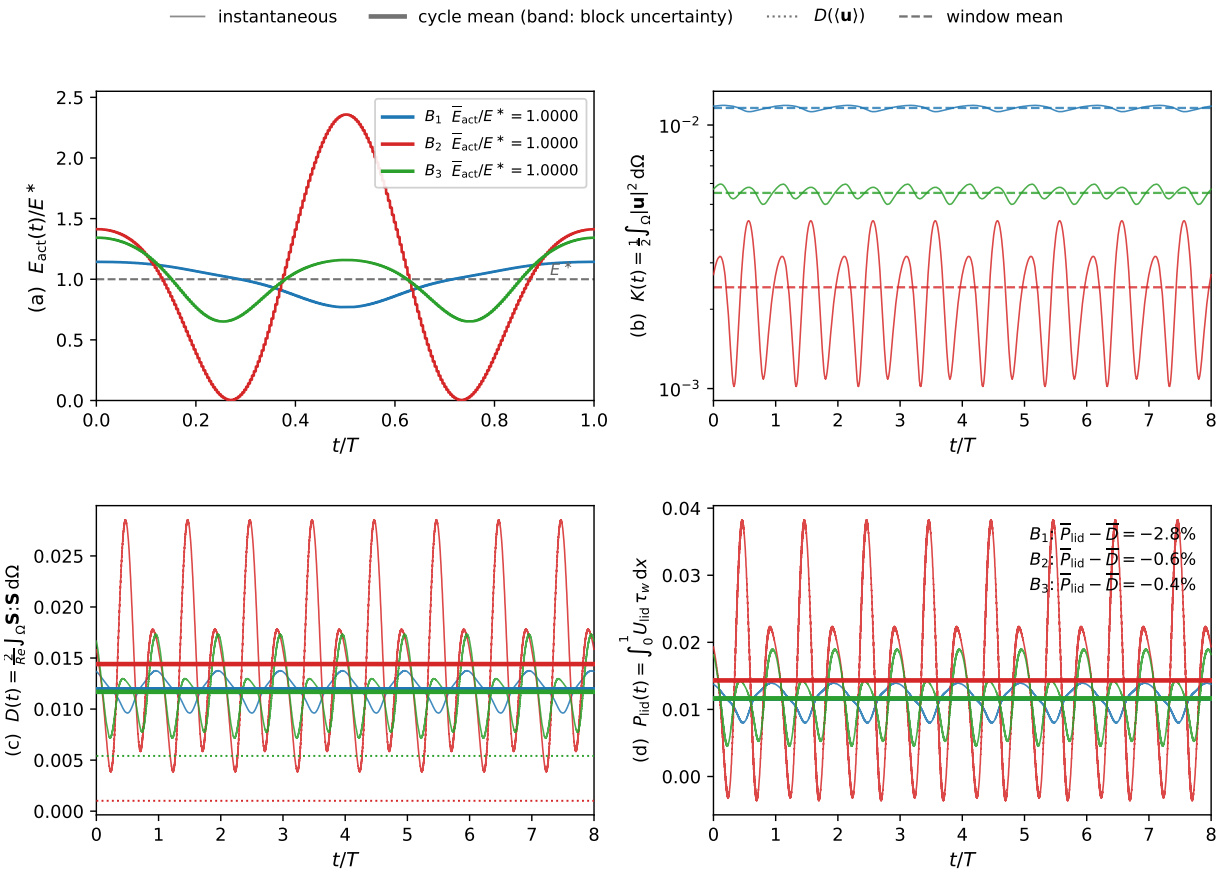


Figure 8: Phase-resolved energetic response of the three branches at $Re = 500$ and $E^* = 0.25$. (a) Instantaneous actuation norm $E_{act}(t)$ over one forced period, normalized by the common cycle-mean target. (b) Kinetic energy $K(t)$ over the first eight of the thirty-two sampled periods (logarithmic ordinate). (c) Viscous dissipation $D(t)$: instantaneous (thin), cycle mean (thick, with the block uncertainty as a band) and, dotted, the dissipation $D(\langle\mathbf{u}\rangle)$ of the time-mean field, which is not substituted for the cycle mean. (d) Lid power $P_{lid}(t)$, positive into the fluid, with the same conventions; the annotations give the residual between the cycle means of the two signals, -2.8, -0.6 and -0.4%, both computed with the wall velocity recovered at the outermost fluid row, the lid node row. Sec. S4.8 compares this estimator with the one built on the prescribed lid velocity, whose residuals are $+3.4$, $+4.6$ and $+4.7\%$, and bounds their difference. The kinetic-energy balance integrates to zero over one period of a periodic state, so the two cycle means must agree; the measured residual is the wall-shear estimator error discussed in the text.

with a few-per-cent estimator uncertainty rather than as an exact closure. This is sufficient for the conclusions drawn here, which rest on the composition of the cycle mean into mean and fluctuating parts — factors of 1.02, 2.2 and 14 — and on a twenty-per-cent separation between the branch totals, not on the third significant digit.

7 Physical interpretation and implications for flow control

7.1 Why the stationary second harmonic increases dissipation

At matched quadratic actuation norm the second harmonic is not a low-dissipation mode, and the reason is geometric rather than spectral. The lid profile $A_2 \sin(2\pi x)$ imposes two opposite lobes of wall velocity, which drive two counter-rotating cells of comparable strength instead of one primary vortex. That reorganization has two competing energetic consequences. It removes the corner velocity jump, which suppresses a singular strain peak that dominates the dissipation of the uniform lid. It also forces the return flow through a wall-to-wall separatrix at the cavity

centre, which raises the strain over the whole interior: the interior mean of $|\mathbf{S}|$ increases by 40% at identical norm. The second effect wins, because the corner layer is thin while the interior is not. The first harmonic removes the same corner peak at almost no interior cost, and is the only stationary profile that is a net reduction. The energetic cost of a spatial actuation mode is thus controlled by the volume over which it raises the strain, not by the peak it removes; a mode that is "smoother" at the wall need not be less dissipative in the bulk.

A purely spectral argument would over-predict that separation, and the size of the error is informative. For a Stokes flow in a half-space beneath a wall that moves tangentially as $A\sin(k\pi x)$, the stream function $\psi = Ay\,e^{-k\pi y}\sin(k\pi x)$ yields a wall shear $\partial u/\partial y|_{y=0} = -2k\pi A\sin(k\pi x)$, so the maximum wall traction is $2\mu k\pi A$ and the dissipation of the layer integrates to $\nu k\pi A^2$, linear in k. Two modes of equal peak amplitude have equal quadratic norm, so the second harmonic should dissipate exactly twice the first. Measured, it dissipates 1.66 times as much at $N = 512$ (0.01185/0.00713), 1.64 at $N = 256$ and 1.59 at $N = 128$ (Table 2): the wall-traction argument gets the direction of the effect, but the observed separation is 17% below the half-space prediction at every resolution. Two effects move the ratio down. The layer depth $(k\pi)^{-1}$ is not small against the cavity height for $k = 1$ or 2, so the side and bottom walls modify the map rather than leaving a half-space; and the interior is inertial ($Re_{\max} = 354$ at matched norm), which moves strain out of the wall layer and into the bulk. The absolute value of the estimate makes the second point sharply: at $\nu = 2 \times 10^{-3}$ and $A = 0.707$ the layer contributes 3.14×10^{-3} to the $k = 1$ dissipation against a measured 7.13×10^{-3}, so a Stokes wall layer accounts for 44% of the cost; the remaining 56% is not captured by the half-space estimate and reflects the combined effects of finite-domain geometry, streamwise gradients and inertia. It should not be read as dissipation located in the cavity core: the estimate covers only the near-wall region it models, and the strain that leaves the layer is redistributed between the interior and the other walls rather than assigned to either. Whether the ratio tends to the Stokes value as Re decreases is a direct test of the inertial explanation, and it is the natural next calculation for the stationary pair.

7.2 Why temporal forcing increases the cycle-mean dissipation

For a periodic control the relevant functional is the cycle mean of a quadratic quantity, and the fluctuations are a first-order contribution to it, not a correction. Equation (8) makes this explicit, and the B_2 branch is its clearest illustration: a mean field that dissipates 1.02×10^{-3} and a cycle mean that dissipates 1.44×10^{-2}, with the mean field holding 25% of the kinetic energy. The physical reason is that the oscillating lid generates a Stokes layer whose wall-normal gradients are large and reverse twice per period; the instantaneous strain is therefore large wherever the instantaneous velocity is large, and the time average of its square cannot be recovered from the average velocity. The same mechanism explains why the lid power of B_2 is negative for 17% of the period: while the Stokes layer spins down, the flow pushes the wall and returns part of the energy that it received earlier. A control designer who evaluates such a state from its mean field will conclude that it is almost frictionless, whereas the cycle average ranks it as the most dissipative of the three branches — by a factor of only 1.2, however, because the two branches whose mean flow survives (B_1 and B_3) pay in the mean what B_2 pays in the fluctuation.

7.3 What branch accessibility does and does not imply

The three branches were not selected by their dissipation. They were selected by a physics-informed optimization whose loss contains a dissipation objective, an actuation-norm constraint and a spatial-variance regularizer, and whose outcome depends on the initialization and on the regularization weight [38]. At $Re = 500$ and $E^* = 0.25$ the stationary second-harmonic structure is one of the most frequently recovered ones, and the full B_1 series is quasi-stationary and reproducible; that is a statement about the variational landscape, and it is confirmed here by the independent solver. It is not a statement about efficiency. At the same actuation norm the

branch most frequently recovered (pooled over the classification campaign $\widehat{P}(B_1) = 42.2\%$ and $\widehat{P}(B_2) = 43.8\%$, i.e. the two are comparable in frequency) is a net dissipation penalty relative to the first harmonic, and the branch with the strongest temporal content, which is also recovered in a reproducible way, is the most dissipative of the three. The operational conclusion is that accessibility and physical performance are different properties, and that a control study which reports only the recovered mode, or only its mean-field dissipation, can invert the energetic ranking of the actuations it compares.

Two limits of this statement must be stated explicitly, because they define its scope. First, the temporal comparison covers one realization per branch at a single Reynolds number ($Re = 500$), on two grids. Each realization is internally converged — the drift of $\overline{D}$ between the two halves of the sampling window is below 1.5×10^{-5} in relative terms and the phase-resolved fields reproduce to better than 2×10^{-3} in relative norm (Sec. 6.5) — but the spread across different initializations of the identification network is not sampled here, so the branch values should be read as properties of the three profiles that were exported, not as class means or as bounds on an entire class. The stationary comparison, by contrast, rests on three grids and two reference lids. The temporal branches B_2 and B_3 contain harmonics up to $n = 4$ in the Fourier basis $\cos(2n\pi t/T)$; at $Re = 500$ the corresponding Stokes layer thickness is $\delta_n = \sqrt{2\nu/(n\pi)} \approx 0.018L$ for $n = 4$ and $0.036L$ for the fundamental, which represents about 4.6 and 9 grid cells at $N = 256$ (Sec. 6.3), and twice that at $N = 512$. Since both branches concentrate 93% or more of their oscillatory energy on the fundamental, this marginal resolution of the higher harmonics is not what sets their cost, and the refined captures quantify how much of the cost is resolution at all: they move $\overline{D}$ by +7.3% (B_2) and +7.1% (B_3), leave the fluctuating shares at 92.8% and 54.8%, and preserve the ordering of the branches, so the residual resolution dependence of the temporal comparison is bounded at the seven-per-cent level, well inside the 20.7–23.5% by which B_2 exceeds the other two branches at both grids. Second, the B_1 temporal content is small but not zero ($f_t = 0.013$), and its cycle mean exceeds its mean-field value by 1.8% at $Re = 500$; the quasi-stationary limit is approached, not attained. The identification campaign already shows that the temporal composition of the optimum changes with Reynolds number: the controls identified at $Re = 1000$ in the preceding study [38] carry a substantially larger temporal energy fraction than B_1 does at $Re = 500$, so the Jensen gap between $\overline{D}$ and $D(\overline{\mathbf{u}})$ is expected to widen with Re. The present three captures hold Re fixed, so they cannot separate that evolution from a pure Reynolds effect; an independent test would require transferring the same control to different Reynolds numbers with consistent normalization, which is beyond the scope of this article. What they do show, at fixed Re and fixed actuation energy, is that the distinction is not merely formal: the ratio $\overline{D}/D(\overline{\mathbf{u}})$ spans 1.02 to 14.2 across the three branches, and that spread is a property of the temporal composition of the control, not of its amplitude. Generalization to three-dimensional cavities, or to actuations with a different spatial support is not implied by the present results.

8 Conclusions

Four statements summarize the study.

1. *The spatial structure of wall actuation reorganizes the cavity and the strain field.* At matched quadratic actuation norm the second spatial harmonic replaces the single primary vortex of the uniform lid by two counter-rotating cells and raises the interior mean strain rate by 40%; the first harmonic removes the corner strain peak at almost no interior cost. The corner singularity of the discontinuous lid is an energetic feature in its own right: it prevents the convergence of the reference dissipation ($p_{\mathrm{obs}} = 0.18$, GCI 131%) and disappears once the corner is regularized ($p_{\mathrm{obs}} = 0.96$, GCI 7.8%).

2. *The ranking of stationary profiles depends on the constraint.* At equal actuation norm, and relative to the grid-convergent regularized reference, whose fine-grid GCI is 7.8%, the

first harmonic reduces the dissipation by 17.7% on the $N = 512$ lattice, while the second harmonic and the mean profile of the identified stationary branch increase it by 36.8% and 44.3%. At equal lid speed the same three profiles are all less dissipative than the uniform lid and the second harmonic is the best of them [38]; the inversion is produced by the norm constraint, which fixes the rms wall velocity but not the peak, and therefore not the effective Reynolds number ($Re_{\max} = 250$ for the uniform lid, 354 for the harmonics, 390 for the B_1 mean profile at $E^* = 0.25$).

3. *The cycle-mean dissipation of a time-dependent actuation is not the dissipation of its mean field, and structurally different actuations can cost the same.* For the lid-oscillatory branch the two differ by a factor 14.2, and the fluctuating part of the strain field carries 92.9% of its cycle-mean cost; for the mixed branch it carries 53.7%, and for the quasi-steady branch only 1.8%. The three branches nevertheless dissipate within 21% and 23% of one another (1.44, 1.19 and 1.17×10^{-2} at a common actuation norm of 0.25): the oscillatory branch is the most expensive not by a wide margin but by a modest factor of 1.2, and the quasi-steady and mixed responses are nearly equal, differing by 2.1% at $N = 256$ and by 0.5% at $N = 512$. The second-Stokes estimate of Sec. 6.3 reproduces the fluctuating dissipation of the lid-oscillatory and mixed branches (B_2 and B_3) to within approximately 5% and 2% from the imposed lid signal alone, and the Reynolds-stress forcing of Sec. 6.4 is spatially consistent with a steady-streaming mechanism, although a closed mean-vorticity budget has not yet been established.

4. *At fixed control effort the dissipation is set by the structure of the actuation, not by the ease with which an optimizer finds it.* The quasi-steady two-lobe structure is one of the most frequently recovered ones and is a net penalty at matched norm; the lid-oscillatory branch is also recovered reproducibly and is the most dissipative of the three. Accessibility and reproducibility describe the variational problem that generated the candidates; the cycle-mean dissipation describes the fluid, and only the second is a performance index.

The practical implication for modal flow control is that a comparison of actuation strategies must state its constraint explicitly. Equal wall speed, equal quadratic norm and equal mechanical power define three different problems and can produce three different rankings; the present study uses the quadratic norm because it is the constraint available to an optimizer, and reports the peak velocity so that the effective Reynolds number of each comparison is visible. The cycle-mean dissipation, not the mean-field value and not the modal spectrum, is the performance index that follows from the energy identity.

Three extensions follow directly. A grid study of the time-dependent branches would establish whether the near-degeneracy of the branch costs and their fluctuation shares are resolution-independent to the same degree as the stationary ranking. Single-harmonic lid excitations at $n = 1, 2, 4$ with identical oscillatory energy would test the $\sqrt{\omega_n}$ dependence of Eq. (15) directly, rather than through the near-degeneracy that the present three spectra produce. Retraining under an explicit cycle-mean norm constraint, rather than rescaling an identified control after the fact, would separate the effect of the constraint from the effect of the identification path. Finally, three-dimensional cavities would introduce spanwise modes and test whether the two-cell second-harmonic state survives when the flow can vary along the span. The physics-informed identification remains, in this programme, the tool that generates the candidate actuations; the fluid-mechanical evaluation is what decides between them.

Data availability

The scripts that regenerate all tables and all figures except Fig. 4, together with the three exported actuation series, their checksum manifest, the solver run logs and the derived records quoted in the text, are available at https://github.com/Ahmedalgeria778/PoF_cavity_dissipation_

study and from the corresponding author on reasonable request; the committed script `scripts/certify_package.py` inventories every figure and table source that the two documents read, names the script that writes each of them and records the compiled page counts. The raw phase-resolved field captures that those scripts read are held separately from the git tree, which is why the repository alone is not a complete data deposit: the six archives of the temporal campaign, `capture_B1_D2_N256.npz`, `capture_B2_D2_N256.npz`, `capture_B3_D2_N256.npz`, `capture_B1_D2_N512.npz`, `capture_B2_D2_N512.npz` and `capture_B3_D2_N512.npz` (28.8, 28.6, 28.7, 111.5, 111.0 and 111.3 MB; 419.90 MB in total), together with the exact-in-time control `capture_B2_D2_N256_exact.npz` (28.24 MB), are supplied as the bundle `PoF_d2_captures_v1.tar`. The committed manifest `results/09_d2_capture/MANIFEST.md` lists every archive with its byte size and SHA-256 checksum, so that a copy obtained from the corresponding author can be verified before the scripts are re-run. The bundle is available from the corresponding author on reasonable request. Fig. 4 is drawn by the committed script `scripts/make_fig_mechanism.py`, but its three panels are rendered from the steady velocity fields of the stationary $N = 512$ campaign (uniform lid, $k = 1$, $k = 2$), raw dumps of 16.0, 16.0 and 13.0 MB; those three archives are held outside the git tree and are supplied on the same terms. The committed manifest `results/10_fig4_stationary_fields/FIELDS_MANIFEST.md` names them (`fields_UNem_0p5_Re500_N512.npz`, `fields_S1em_0p707107_Re500_N512.npz` and `fields_B1em_0p707107_Re500_N512.npz`) with their byte sizes and SHA-256 checksums, supplies them as the bundle `PoF_fig4_fields_v1.tar` and verifies that they reproduce the peak and interior-mean strain-rate magnitudes quoted in Sec. 5.3.

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Author declarations

Conflict of interest

The authors have no conflicts to disclose.

Author contributions

Beniaiche Ahmed: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead). **Larabi Abderrahim:** Conceptualization (supporting); Formal analysis (supporting); Methodology (supporting); Supervision (lead); Writing – review and editing (supporting). **Belkadi Mbarek:** Formal analysis (supporting); Supervision (supporting); Writing – review and editing (supporting).

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